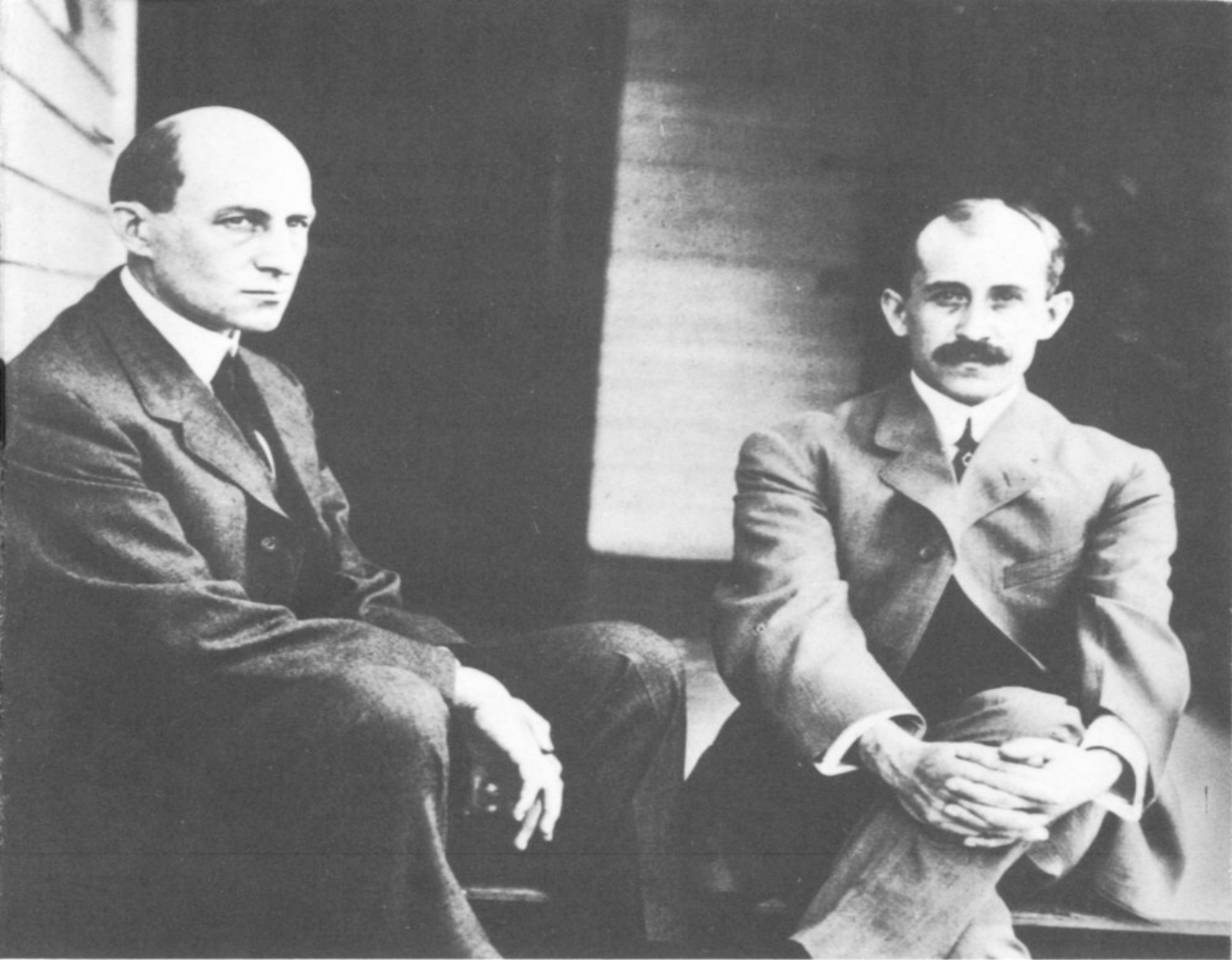
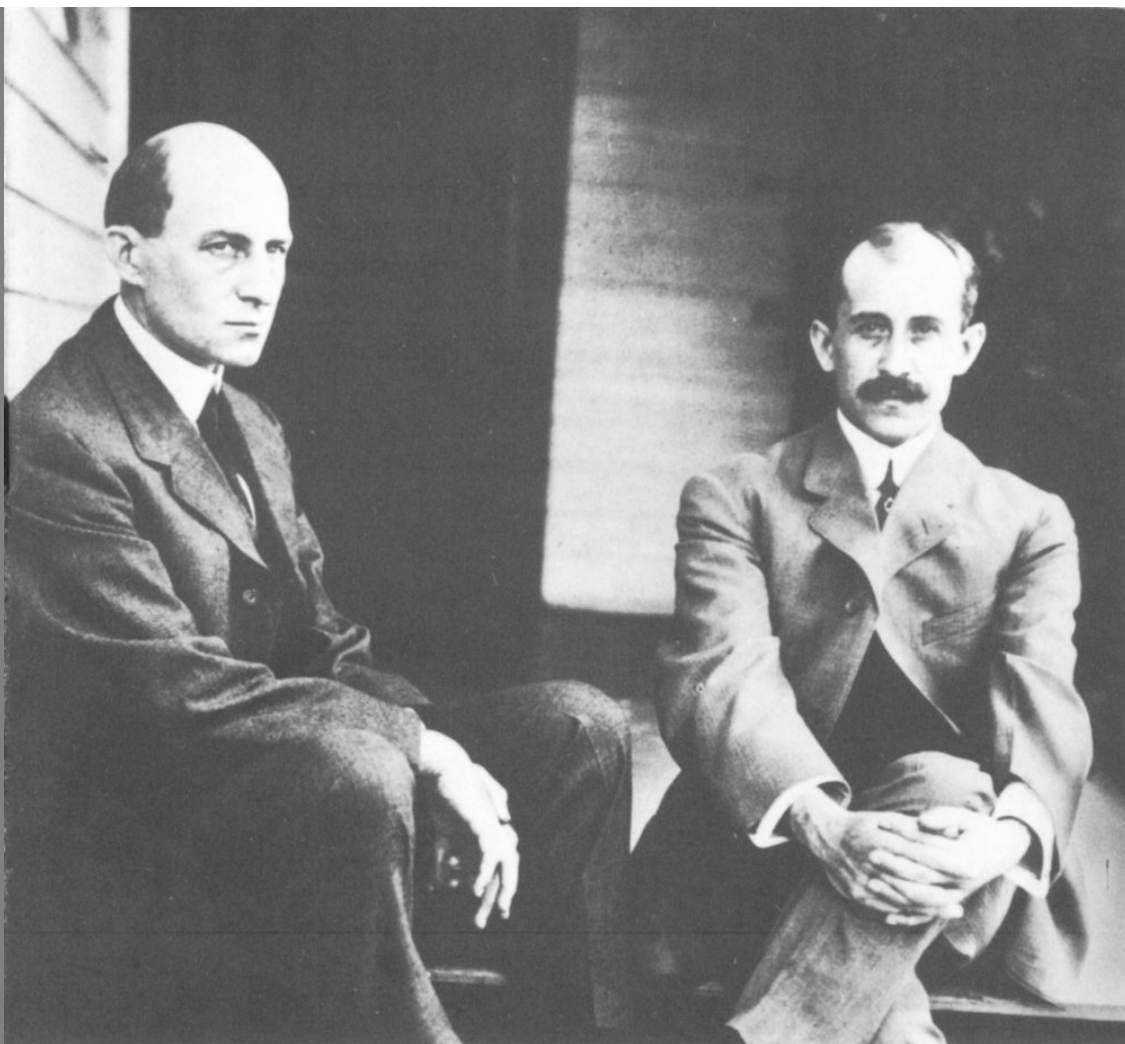




The

WRIGHT BROTHERS

"As scientists, they discovered the secret of flight. As inventors, builders and flyers, they brought aviation to the world."



The
WRIGHT BROTHERS

"As scientists, they discovered the secret of flight. As inventors, builders and flyers, they brought aviation to the world."

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*** START OF THE PROJECT GUTENBERG EBOOK THE WRIGHT
BROTHERS ***

The **WRIGHT BROTHERS**

“As scientists, they discovered the secret of flight. As inventors, builders and flyers, they brought aviation to the world.”

On December 17, 1903, at Kitty Hawk, North Carolina, the first power-driven heavier-than-air machine ever to achieve sustained flight rose from its starting track and in 12 seconds soared through the air for a distance of 120 feet. Short as this flight was, it nevertheless marked the beginning of man's conquest of the air. Orville Wright was at the controls; Wilbur Wright balanced the machine at the take-off. This picture records for posterity an epochal event witnessed by just seven men, the Wright brothers themselves and five others who, more than they knew, stood that day on the threshold of history.



The first flight at Kitty Hawk, N. C., December 17, 1903.

1



The Age of Flight, with its miracle of service, began in an obscure little bicycle shop in Dayton, Ohio. Here two devoted brothers, working amid tires, wheels and air pumps, dreamed that man could fly in a heavier-than-air machine. The names of these two brothers who wrote themselves indelibly into history, were Wilbur and Orville Wright.

The story of the Wright brothers is an inspiring narrative of success. Wilbur and Orville combined to a rare degree the searching intelligence of the scientist, the ability to visualize of the inventor, and the practical craftsmanship of the builder. In addition they had great personal courage.

The Wright brothers were by no means the first who sought the secret of flight. Particularly in Europe, able men had delved deep and risked much in the effort to fly like a bird. Certain theories of aerodynamics had been developed and were generally accepted as accurate. One of the major setbacks to the hopes of the Wrights was the discovery, through their own experiments, that these previously accepted theories were incorrect.

This meant that they had to start from the beginning and develop their own tables of air pressures. Two developments of the Wrights made it possible to build an aeroplane that would fly. One was a crude wind tunnel and the other was an ingenious set of balances made out of old hack saw blades and bicycle spokes. With these comparatively crude instruments, they compiled data which made flight possible.

In the months and years following their first flights, the Wrights were acclaimed by nations and by men. They knew success in the fullest measure. But probably no subsequent achievement quite equaled the thrill which must have been theirs when they were able to send to their father and sister that now famous message:

“Success four flights Thursday morning all against twenty-one-mile wind started from level with engine power alone average speed through air thirty-one miles longest 59 seconds inform press home Christmas.”

An idea is born



The shop of the Wright Cycle Company on West Third Street in Dayton ... birthplace of the aeroplane.

The Wright brothers sprang from pioneers who settled Dayton when the Ohio country was young. Their father, the Reverend Milton Wright, became a bishop of the United Brethren Church. His vocation necessitated frequent changes of residence. Thus it came about that Wilbur was born April 16, 1867, on a farm eight miles from Newcastle, Indiana, while Orville was born in a house at 7 Hawthorn Street in Dayton. This house was the Wright home for more than forty years.

From earliest childhood, the boys were mechanically minded. They had both the inclination and the aptitude for creative work. The pioneering urge and the gift of original thinking were theirs.

WEST SIDE NEWS

No. 19.

DAYTON, OHIO, JULY 20, 1889.

Vol. 1.

West Side News.

PUBLISHED WEEKLY.

Walter Wright Editor
Orville Wright Publisher

TERMS:—Quarter of year, twenty cents
Six months, two cents.

103 WEST THIRD STREET.
DAYTON, OHIO.

Seeing a Bird's Nest.

Edward Corlin, the inventor of the Corlin engine, in building an addition to his factory, while laying the foundation found it necessary to remove a ledge by blasting. The workmen had been employed, the materials provided, and the blasting begun. The next morning Mr. Corlin passed the place where work was proceeding, when the foreman in charge, knowing his interest in pretty things, called him.

"Here, here, Mr. Corlin," said he; here is a bird's nest that we have found, and that's got to go. He showed the manufacturer a robin sitting upon a nest that had been built fast and snug in a crevice of the rock among some bushes. The bird flew off her nest as the man came near, and showed five blue eggs, that looked as if they had just been laid.

"Can we move the nest somewhere else?" asked Mr. Corlin. "I'm afraid not, sir. We'd tear it to pieces getting it out, and it isn't at all likely you could get the bird to go to sitting again anywhere else. We've got to go on, as we may as well rip it out, and throw the eggs away. Let her bring out her brood right here."

"But we'll have to stop the work on the building!"

"Let us stop it, then."

And so orders were given that operations on the addition should be suspended. They were suspended; and the hands stood still, drawing their pay for doing nothing, or next to nothing, while the robin sat on her nest with an air of great consequence and jealous attention to business, and had her food brought her by her mate, and at last hatched her brood. And then there were three weeks more, at the least, before the young ones could fly.

Mr. Corlin visited the nest frequently, not with any impatience to have the robin and the young

ones out of the way, but with a genuine interest in their growth. The old birds had all the time they wanted; and when at last they had stately helpers over the edge of the nest, and they showed themselves able to get about on their own hook, orders were given to resume the building operations; and the dull boom of the gunpowder tearing the rocks apart was heard where the birds had peeped.—*Boston Transcript.*

"Dat Bargain."

A little colored boy, not over intelligent but full of soul, called at a house and requested the privilege of removing three barrels of rubbish for the sum of twenty cents. The request was granted, and the next morning the young African appeared with two comrades and a small push-cart.

When the sales were about half removed, a halt was ordered, and the three trembling with a sense of their wrongs.

"Lady, didn't you make dat bargain wid me fur twenty cents?" demanded the boss.

"Yes," demanded one of the assistants. "I let him hab de use ob my kyart. (tra'n't he ter pay me fur it? He says he not gwine ter pay me er cent!"

"N, I help too!" put in Number Three, a handsome little mulatto. The question was finally left to the arbitration of a plumber, whose judgment in financial affairs made him the Solomon of the neighborhood.

"Pay the boss ten cents, and each of the workmen five," was his stentorian order.

"Den I strike, 'n' not 'low de oder feller de use ob my push-kyart." "N' I not help him tote dem barrels!" said Number Three. So the case was appealed.

The boss, finding that his average was likely to lose him every thing, compromised on seven cents for himself, seven cents for the owner of the push-cart, and six cents for the third member; but his disgust was expressed as follows: "Dese aint noffin' ter be made outen dese contracts, nobow. I've done totin' ashes!"

Astride a Bear.

Among the incidents of the early history of Camden, Maine, which have been handed down,

none is more interesting to the young people than an adventure of Leonard Metcalf, one of the first settlers of the town. Metcalf, a young man named Webber, and a young man named Richards, were out one day on a moose hunt. The only gun was carried by Richards; the two others went armed with a hatchet each.

As the company divided, Richards kept to the base of Mt. Battie, while the others ascended it and kept the same course. Mr. Richards' two dogs had scoured up an old bear on the western slope of the mountain, and, after shooting it, Mr. Richards went in search of his companions to apprise them of his success.

In the meantime, Metcalf and Webber had found the lair, where the two cubs, of considerable size, were secreted. They tried to drive them out of the den, by building a fire and smoking the entrance.

As one of the cubs essayed to come out, Metcalf made a thrust at him with his jack-knife, having carefully laid his hatchet down. The young bear then rushed out of the den when Metcalf, full of eager excitement to capture it, seized the cub by the ears and held on and called to Webber to come and help him.

"Knock him with your hatchet! knock him down, quick!" exclaimed Metcalf, struggling to keep his hold on the cub's ears.

But Webber was not quick enough. He was either afraid, or enjoyed the fun of seeing Metcalf jumping about, and so did not hurry.

Metcalf saw that he could not hold the bear much longer, and, with the rush of youth, jumped astride the animal which dashed with fright down the mountain-side at the imminent risk of the young man's life.

Firmly grasping the bear's neck with one hand, Metcalf clung on, and tried from time to time to cut the animal's throat with his knife, which he held in the other; but the blade kept shutting up, so that his attempts to dispatch him were in vain.

After thus riding down nearly to the base of the mountain, tearing his clothes, and scratching his legs badly by the bushes and crust of snow, he was met by Richards, who fortunately approached and was able to knock the bear in the head, and thus rescue Metcalf from his dangerous situation.

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Physician's prescriptions carefully compounded.

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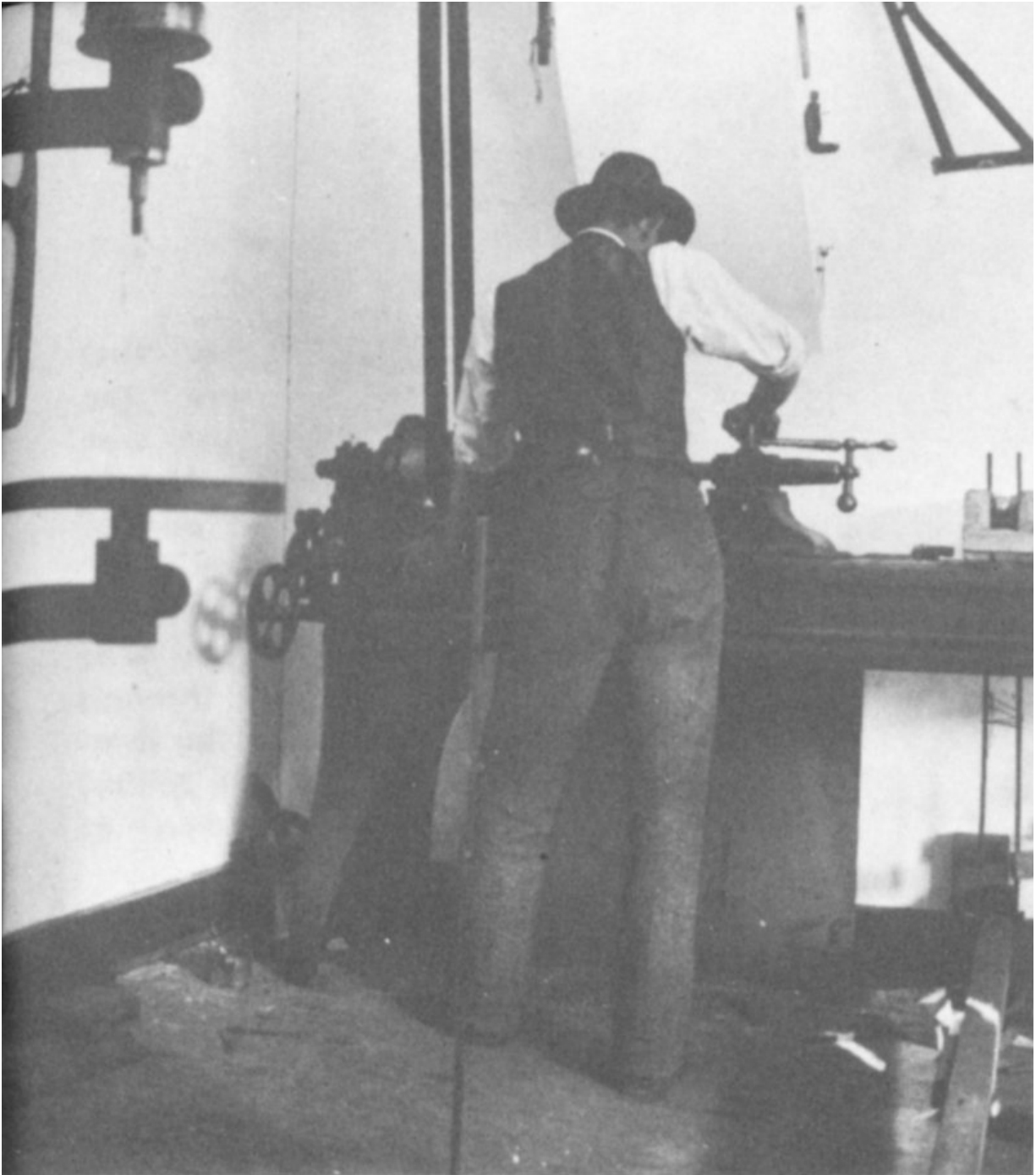
WELL T. SILVER

HEADQUARTERS FOR
Chop feed, Crushed corn;
Steam dried meal.

One Square West River Bridge.
PRUDEN'S FAIR.
Refrigerators, Ice Chests,
Hammocks, Baby Carriages,
House Ball Goods,
House Furnishing Goods, etc.
PRUDEN'S FAIR,
4 and 6 E. 4th St., opp. post-office.

An issue of the "West Side News," an early Wright venture.

One day the Bishop came home from a short trip, bringing the children a present. He held something in his hands and then tossed it toward them. It was a toy helicopter. Instead of flopping to the floor, it ascended to the ceiling where it fluttered before it fell. That helicopter set up a milepost in the lives of the Wright boys. The idea of their future conquest of the air, in all likelihood, was born then and there. 3



Wilbur Wright in the bicycle shop, 1897.



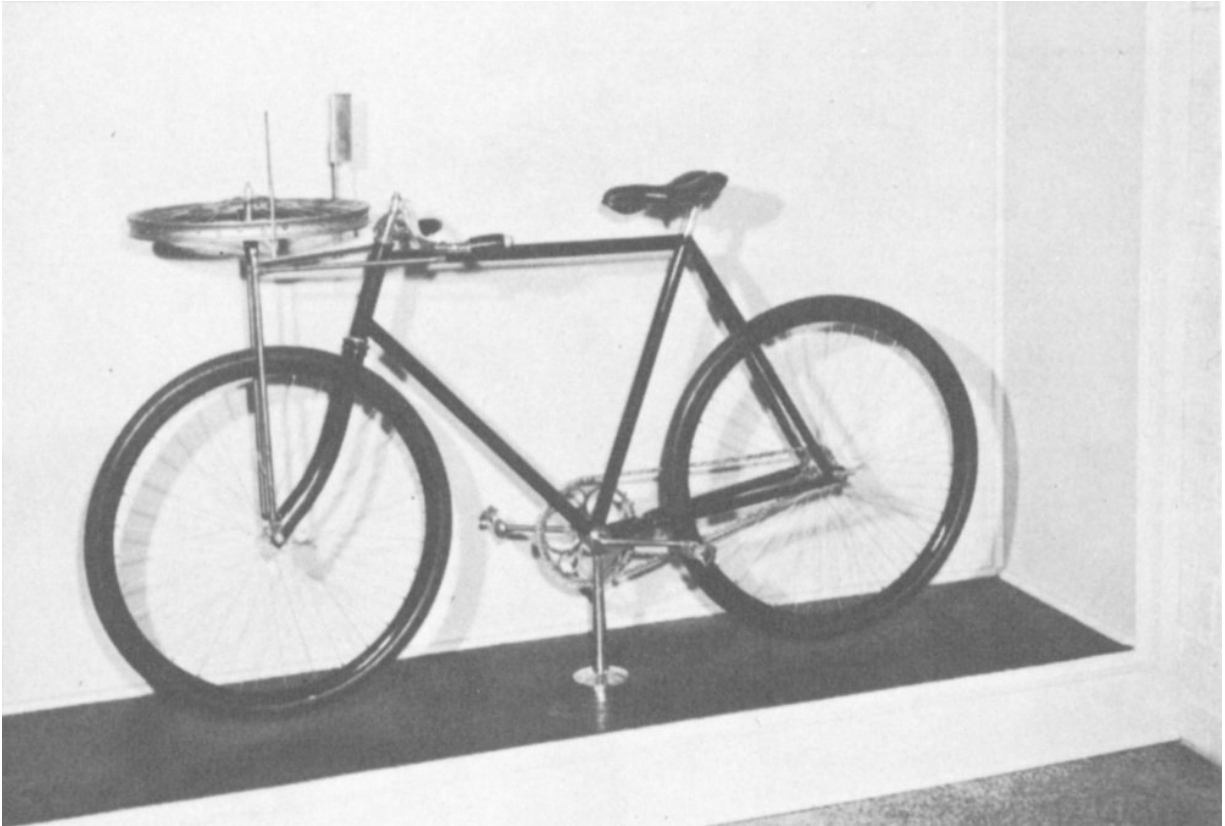
Orville Wright, in white shirt, at work in shop.

At an early age they began to fly kites. They became interested successively in wood cuts, printing and photography. The urge for invention was strong in them. Wilbur got a job folding the entire issue of an eight-page church paper. When he found the handwork tiring and tedious, he designed and built a machine that did the folding.



The house on Hawthorn Street, home of the Wrights for 40 years and now re-erected in Greenfield Village in Dearborn, Michigan.

Orville was no less enterprising. When he was 15, he entered into a partnership with Ed Sines, a neighbor boy, and launched the printing firm of Sines and Wright. The plant was located in a corner of the Sines kitchen. One of their first ventures was to print a little paper called “The Midget.”



One of the Wrights' first efforts to measure the effect of air pressure was this horizontal bicycle wheel mounted on one of their own bicycles and equipped with two metal vanes. This bicycle was placed in the Park through the co-operation of the family of the late Frank Miller, former Superintendent of Dayton Schools.

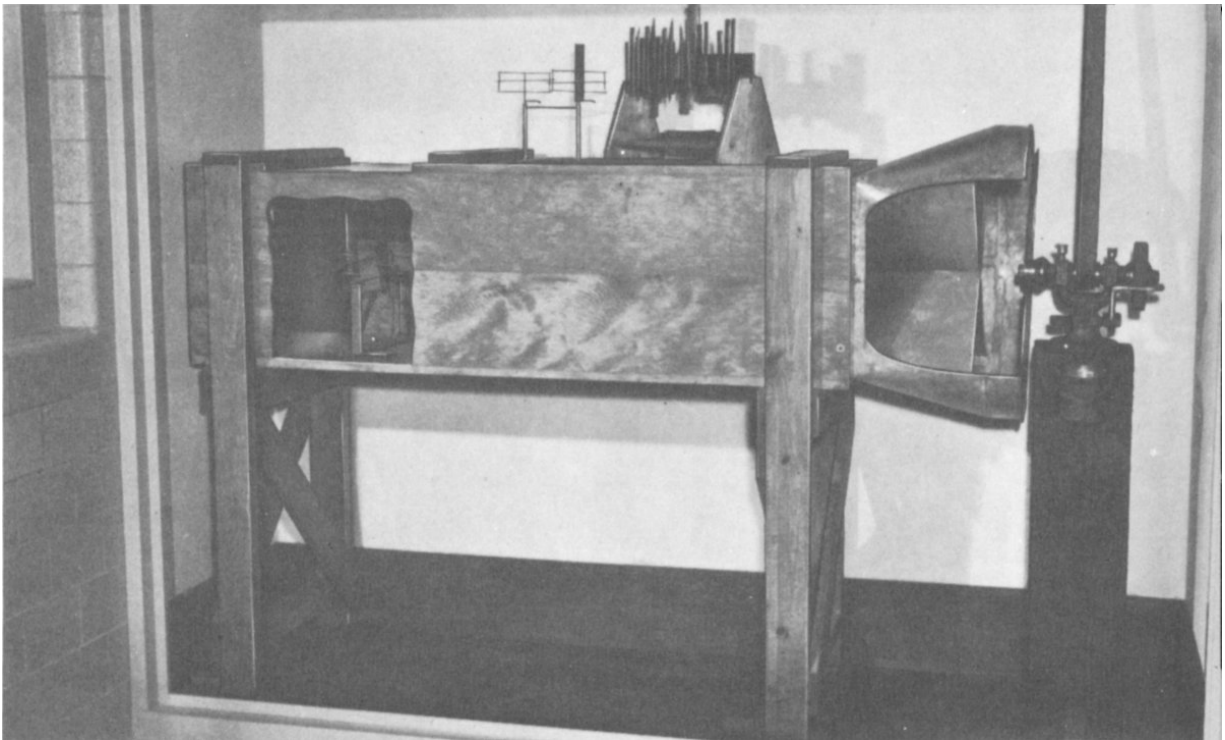
Later Orville started a weekly newspaper called the "West Side News." Wilbur joined him as an editorial writer. These publications and others which followed were printed on a press which the Wright boys designed and built.

In 1892 came the enterprise that was to provide the setting for, and the approach to, the supreme adventure with which the names of the Wright brothers are associated. The boys became absorbed in bicycles. Orville became interested in track racing and participated in several events. In their enthusiasm the boys decided to go into the bicycle business. After embarking on bicycle selling they discovered they must have a repair shop. Punctures provided the bulk of their business, with free air as a side issue. The first shop of what became the Wright Cycle Company was at 1005 West Third Street.

Business increased to such an extent that the Wrights moved to South Williams Street. Here they began to manufacture bicycles. Their first model was called the Van Cleve, named after one of their pioneer Dayton ancestors. Continued expansion of the business necessitated a move to 1127 West Third Street. This was the shop linked with the birth and development of aviation. It was here that Wilbur and Orville not only dreamed of flying but practically built the first plane.

A hint of what the future had in store came one day when the brothers were discussing what was then the new-fangled horseless carriage. Since it was an original idea, it appealed to them. Orville suggested that they might engage in the automobile business. “No,” replied Wilbur, “you’d be tackling the impossible. Why, it would be easier to build a flying machine.”

5

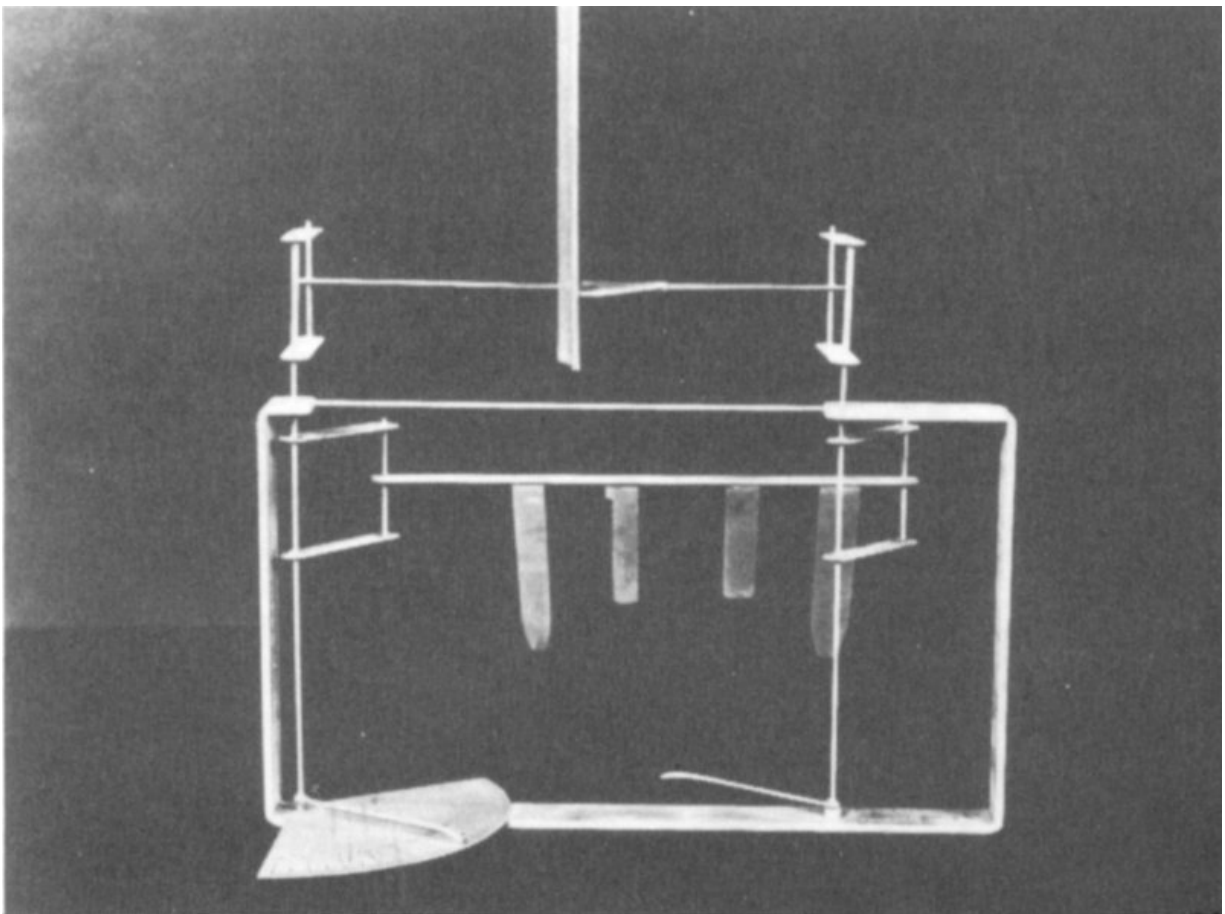


A replica of the Wrights' original wind tunnel which secured its pressure from a fan mounted on the shaft of an old grinding wheel.

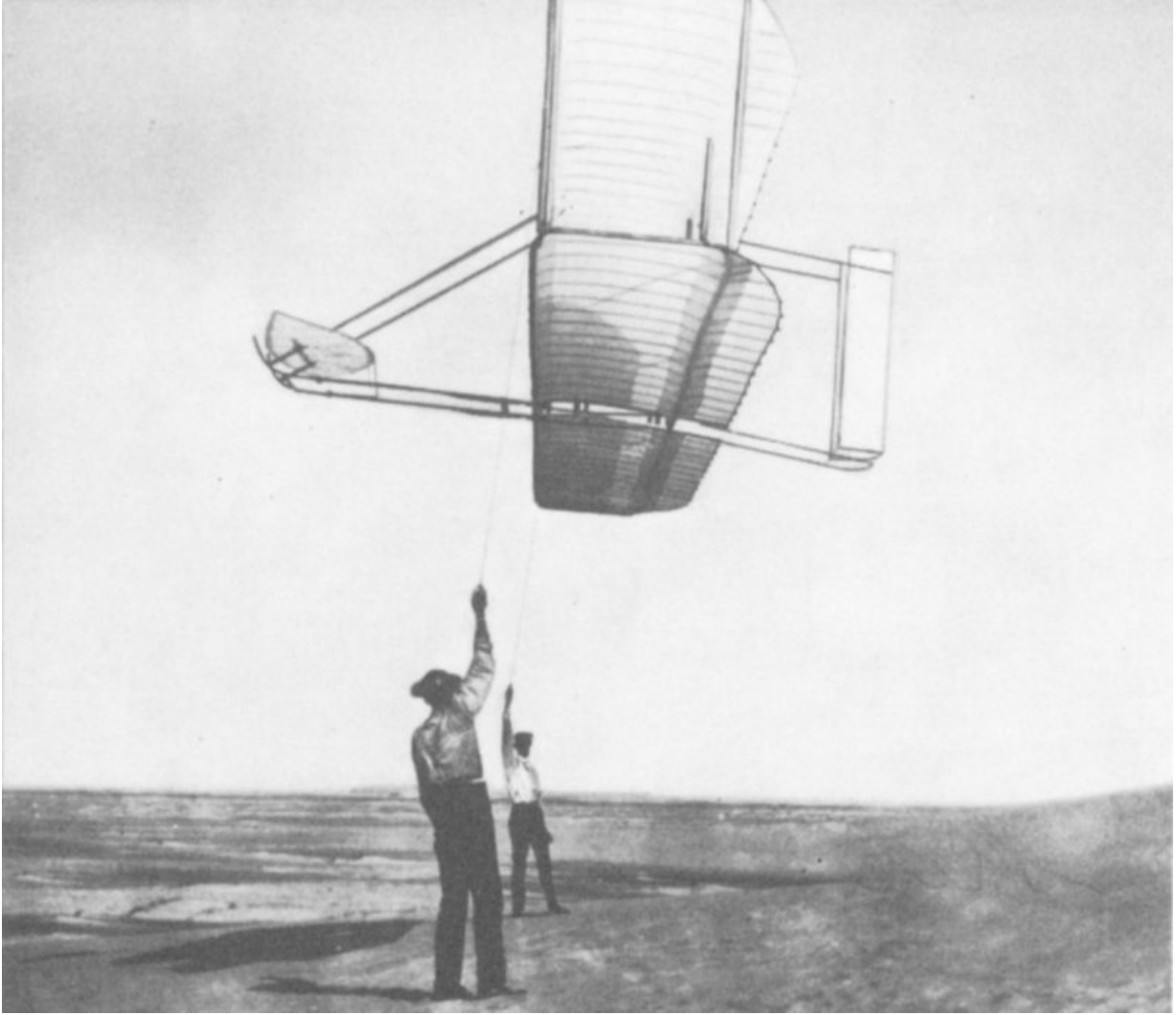
The first active interest in flying that the Wrights displayed developed in 1895 when they read about the glider experiments being carried out by Otto

Lilienthal in Germany. They now began to read everything they could lay hands on that bore on the attempts of man to fly, going back to the days of the great Leonardo da Vinci. They wrote to the Smithsonian Institution for a list of books on the subject. The germ of flying now entered their systems, never to be eradicated.

The Wrights went thoroughly into the problem of gliders. After Lilienthal had been killed while gliding, the brothers discovered that neither he nor any other man who glided had an adequate method of insuring lateral balance. In seeking the solution to this problem, Orville worked out a theory for the operation to vary the inclination of sections of the wings, thereby obtaining force for restoring balance. Thus he hit upon a fundamental principle which became a claim in the original Wright patent.



One of the most valued possessions of the Wrights, a balance made of hacksaw blades. With this balance they evolved their own tables of air pressure which eventually enabled them to fly. The original balance is in Franklin Institute, Philadelphia; this replica is in Wright Hall, Carillon Park.



Many glider flights at Kitty Hawk preceded the first attempt to fly in a power-driven plane. Here the Wrights are flying a glider as a kite, controlling it from the ground. Later flights were made in man-carrying gliders.

The brothers now began to study wing structure, but hit upon many difficulties. A simple incident set them on the right track. In selling a customer an inner tube for a tire, Wilbur had taken the tube from the pasteboard box and was idly twisting the box back and forth as he talked to the customer. In doing so he noticed that although the vertical sides remained rigid at the ends, the top and bottom sides could be twisted so that they made different angles at the opposite ends. He immediately wondered why the wings of a gliding machine could not be warped from one end to

the other in this same way. In this way the wings could be put at a greater angle at one side than the other and there would be no structural weakness. Wilbur explained the plan to Orville and it seemed so satisfactory that they adopted it for their gliders.

The Wrights were now glider-conscious. They built a bi-plane kite with a new system of controls. In 1900 the brothers constructed a man-carrying glider. In order to get practice in operation, they decided to fly it first as a kite. For kite flying they required flat, open country; and for gliding, sand hills free from trees or shrubs were necessary. Favorable winds were also needed.

From reports received from the Weather Bureau in Washington, the Wrights learned that a place named Kitty Hawk in North Carolina seemed to meet all requirements. So they wrote to the man in charge of the weather station there for further information. On his and other data, the brothers came to the conclusion that Kitty Hawk was suitable for experiments. What was then a tiny spot on the map was to become, in time, a center of world interest.

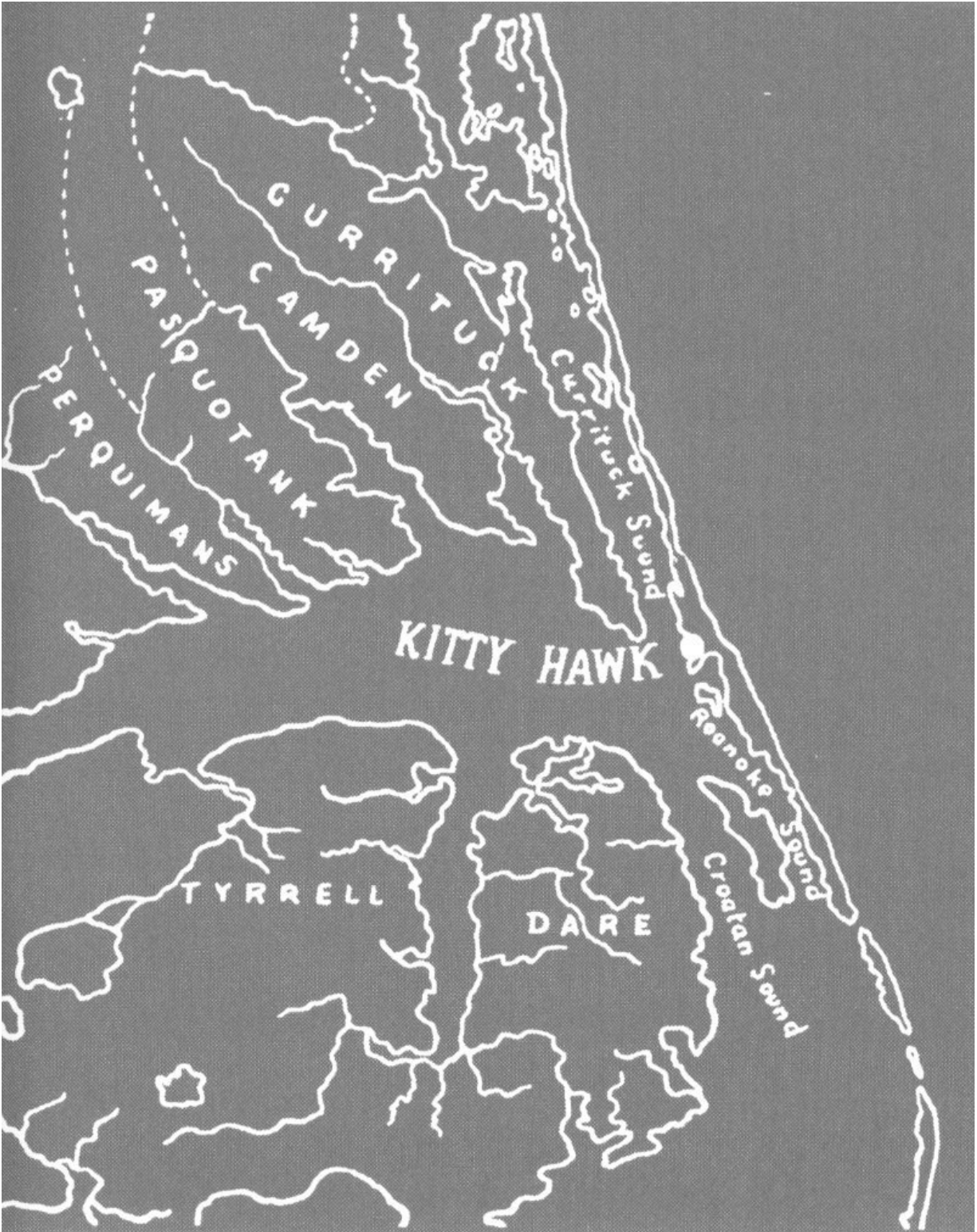
by noon and got the machines out on the trucks ~~by~~ front of the building ready for a trial from the level. The wind was gradually dying and the distance we were ready was being only about 4 to 5 miles per. sec. After waiting some 4 hours to see whether it would breeze up, again we took the machines in.

Thursday, Dec. 17th

When we got up a wind of between 20 and 25 miles was blowing from the north. We got the machines out early and put out the signal in the middle of the station. Before we were quite ready, John T. Daniels, W. S. Douglas, A. D. Etheridge, W. C. Brinkley of Maules, and Johnny Moore, of Nags Head arrived. After

running the engines and propellers a few minutes to get them in working order, I got on the machine at 10:35 for the first trial. The wind according to our anemometer, at that time was blowing a little over 20 miles (corrected) 27 miles according to the government anemometer at Kitty Hawk. In so doing the rope the machines started off measuring in speed to probably 7 or 8 miles. The machines left from the truck just as it was entering the fourth rail. Mr. Daniels took a picture just as it left the trucks. I found the control of the front rudder quite difficult on account of its being balanced too near the center and thus had a tendency to turn itself when started so that the rudder was turned too far on one side and then too

Diary of Orville Wright, showing page recording the first successful flight.



Map of Kitty Hawk area.

Man conquers the air

The spot chosen for the Wrights' experiments was located on a long strip of sandy beach separated from the mainland of North Carolina. At one point was the Kitty Hawk Life Saving Station and alongside, a government weather station. A mile back from the ocean was the tiny village of Kitty Hawk. Four miles south was the Kill Devil Life Saving Station. Altogether, it was a dreary and uninviting area but one where history was to be made.

The Wrights' experiments at Kitty Hawk covered two periods. In 1900 they began flying gliders. Winds proved to be unsatisfactory with the result that the experiments of this year fell far below expectations. They were back in 1901 with a much larger glider. From this model they learned that large surfaces could be controlled almost as easily as smaller ones, provided the control was by manipulation of the surfaces themselves instead of the movements of the operator's body. In their glider experiments of 1901 they broke all records for distance in gliding.

Air lift still troubled the brothers, so Orville rigged up a small wind tunnel made out of an old starch box. Within the box was a balance, the main feature of which was a metal rod that pivoted like a weather vane. The starch box experiment led to the design of the more scientific wind tunnel shown on [Page 5](#).



Newspaper comments on the early efforts of the Wright brothers.

In the third glider trials in 1902 the brothers put all their new knowledge to the test with good results. One new feature was a “tail.” The idea of making this tail movable led to the system of control generally used today—the independent control of aileron and rudder. The third series of glider flights was highly successful.

The Wright brothers were now convinced they could build a successful power flyer. One of the first requirements was an engine to produce at least eight horsepower and weigh not more than 20 pounds per horsepower. Unable to obtain such an engine, they built one themselves. The plane now took shape with wings having a total span of a little over 40 feet with the

upper and lower wings six feet apart. Total weight of the plane was 750 pounds.

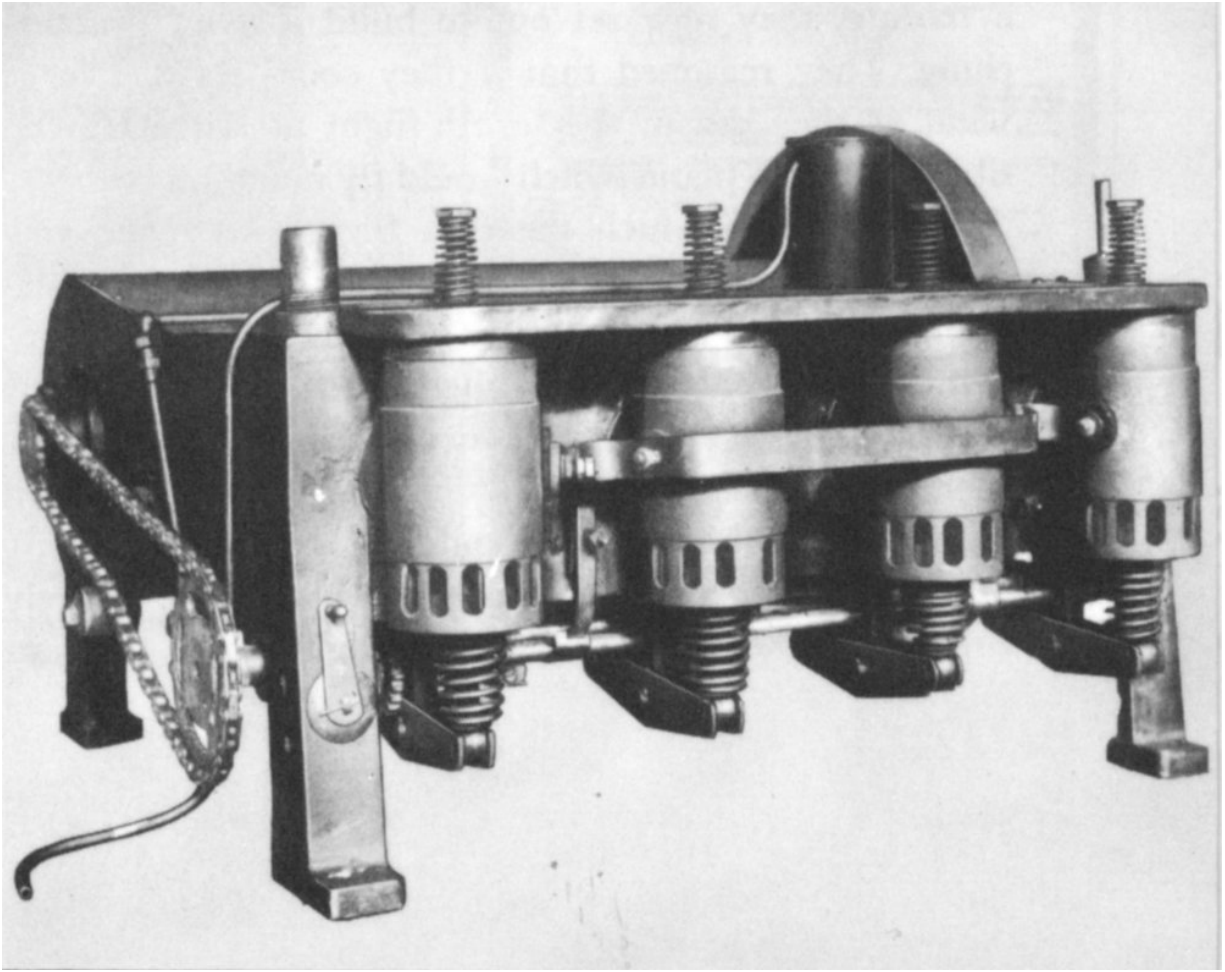
Although the plane was assembled in Kitty Hawk by September 23, 1903, weather and various mechanical mishaps postponed the day of trial until December 14. On the toss of a coin, Wilbur won the right to make the first trial. The machine climbed a few feet, stalled and fell. Several parts 9 were broken, requiring two days for repairs. There were other minor delays and then came the fateful day of December 17.

This time Orville was the pilot. The few spectators stood silently by, little realizing that they were participating in an event that would be “forever known.” Orville lay flat in the pilot’s place with Wilbur running alongside, a hand on a wing, until the machine left the rail. This, in the words of one of the historians of the flight, is what happened:

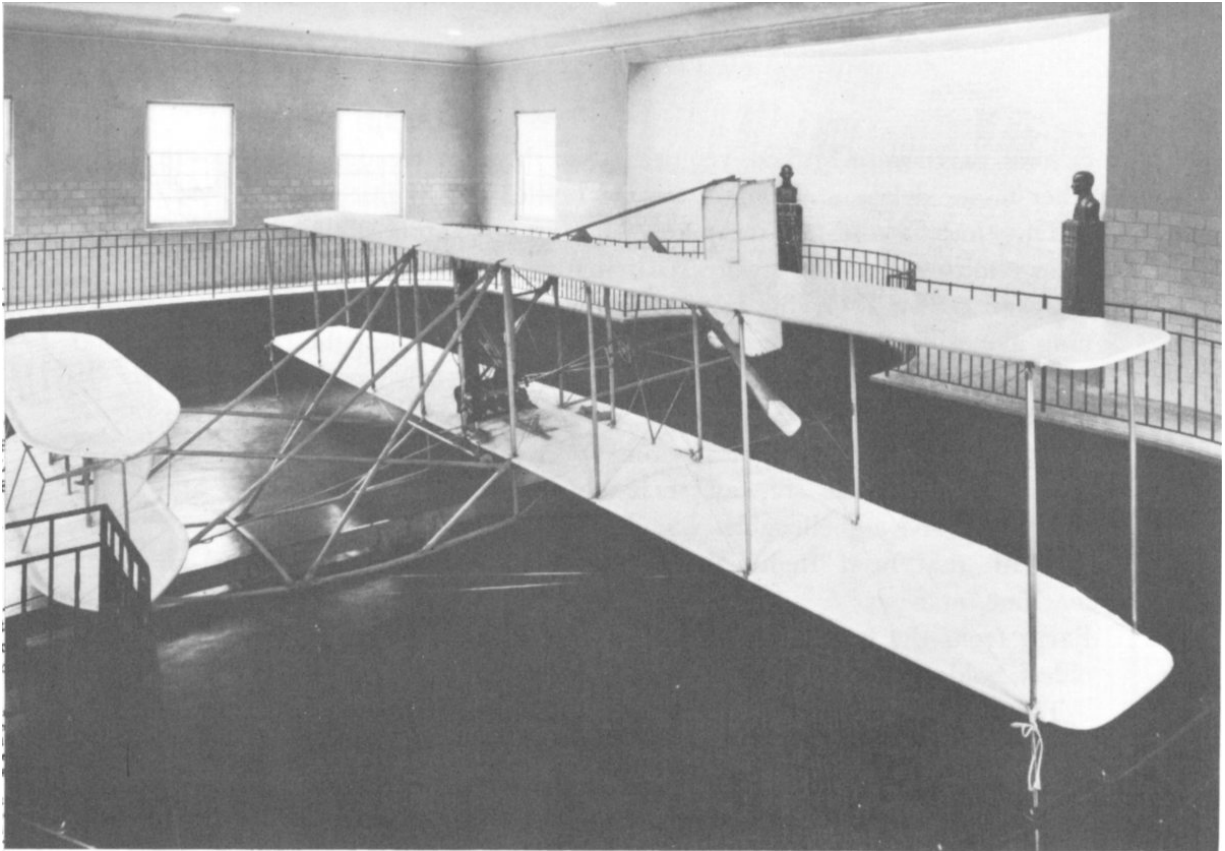
“Signals that all was in readiness were exchanged. The motor turned, the propellers whirled, a restraining wire was released; the machine rolled along a crude runway, then took off under its own power and flew for twelve unbelievable seconds for 120 incredible feet.

“With that brief flight, the first ever made by a heavier-than-air machine, man was freed from the bonds that held him close to Mother Earth from the beginning of time, and glimpsed the realization of his oldest, boldest dream ... the conquest of the air.”

The moment when that homemade plane rose from the ground was akin to others that heralded epochs in the progress of mankind. Crude as it was, that first plane represented an almost incredible amount of preparation. Gliders had been designed, constructed and flown to gain technical data and piloting technique; a satisfactory system of control had been discovered; a wind tunnel and balance had been built to amplify flight data; an aircraft engine sufficiently light in weight had been developed; and finally an aeroplane had been designed and built. All these things were accomplished in about three years. As one challenge followed another, the Wrights met them all and from their first flight went on to the further development of their invention.



The engines used in the first Wright planes were built by Orville and Wilbur and had four cylinders. This is the original engine from the 1903 plane.



The restored 1905 Wright plane in Wright Hall.

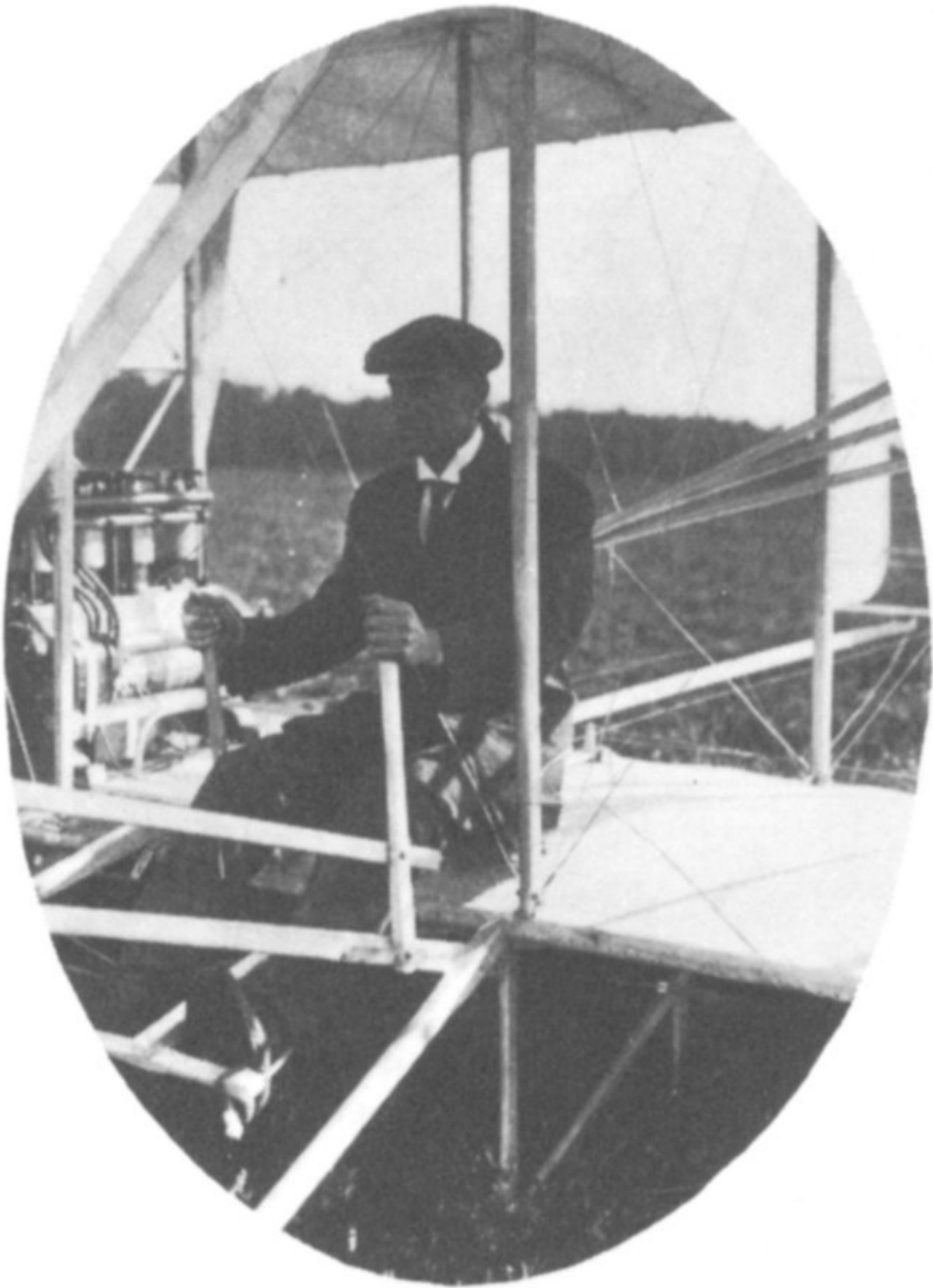
The Wrights learn to fly

Although the first flight took place at Kitty Hawk, the Wrights themselves always said that they really learned to fly on Huffman Prairie east of Dayton on the present site of Wright-Patterson Field. Having proved that they could fly even if for a maximum of less than a minute, they now set out to build a more practical and useful machine. They reasoned that if they could fly 852 feet against a 20-mile wind as they did in the fourth flight at Kitty Hawk, it should be possible to build a plane which would fly much farther.

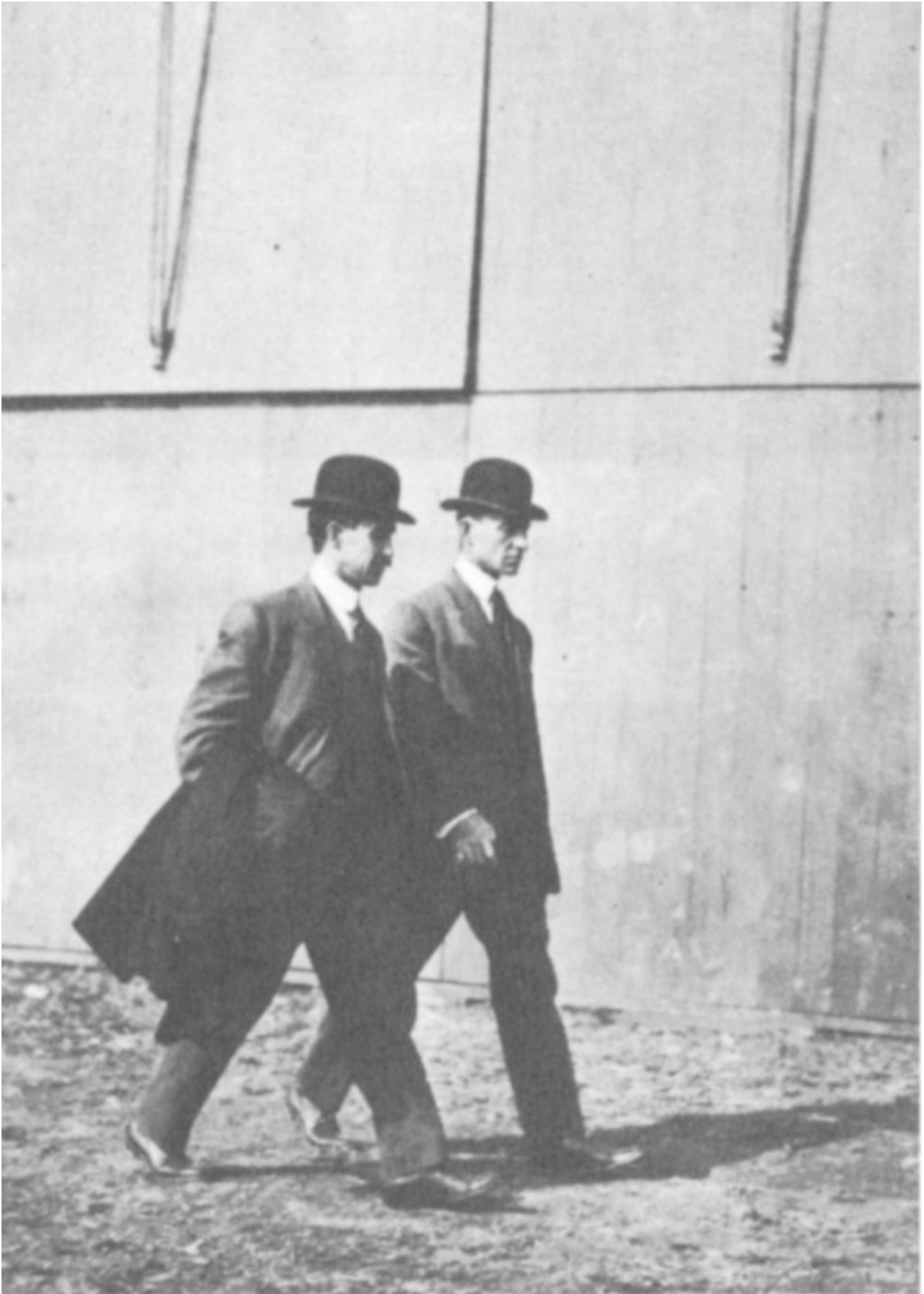
The plane in which the first flight was made was called the Kitty Hawk. Construction of its immediate successor began in January, 1904. It was much the same as the one flown at Kitty Hawk but there were a number of changes and the construction was more sturdy throughout. This plane was equipped with an entirely new engine. Because of a shortage of spruce in Dayton they changed to white pine for spar construction, thinking it would be equally good. However, the pine broke in actual use and the wings had to be entirely rebuilt.



Wilbur Wright during the first demonstrations of the plane in Europe.



Wilbur at the controls during a flight in France.



Orville and Wilbur Wright, modest men whose achievements made history.

821,393

THE UNITED STATES OF AMERICA

TO ALL TO WHOM THESE PRESENTS SHALL COME

Whereas Orville Wright and Wilbur Wright,
of Dayton, Ohio.

HAVING APPLIED TO THE COMMISSIONER OF PATENTS A PETITION PRAYING
FOR THE GRANT OF LETTERS PATENT FOR AN ALLEGED NEW AND USEFUL IMPROVEMENT IN

Flying-Machines.

A DESCRIPTION OF WHICH INVENTION IS CONTAINED IN THE SPECIFICATION OF WHICH
A COPY IS HEREBY ANNEXED AND MADE A PART HEREOF, AND SATISFIED WITH
THE VARIOUS REQUIREMENTS OF LAW IN SUCH CASES MADE AND FURNISHED, AND

Whereas UPON THE EXAMINATION MADE THE SAID CLAIMANT IS DEEMED
TO BE JUSTLY ENTITLED TO A PATENT UNDER THE LAW.

NOW THEREFORE THESE LETTERS PATENT ARE TO GRANT UNTO THE SAID

Orville Wright and Wilbur Wright their heirs and assigns
FOR THE TERM OF SEVENTEEN YEARS FROM THE TWENTY-SECOND DAY OF
May
ONE THOUSAND NINE HUNDRED AND SIX.

THE EXCLUSIVE RIGHT TO MAKE, USE, AND VEND THE SAID INVENTION THROUGHOUT THE
UNITED STATES AND THE TERRITORIES THEREOF.

In testimony whereof I have hereunto set my
hand and caused the seal of the Patent Office
to be affixed at the City of Washington
this twenty-second day of May
A.D. 1906
and of our Lord one thousand nine
hundred and six
and of the
Independence of the United States of America
the hundred and thirtieth

J. L. Bell

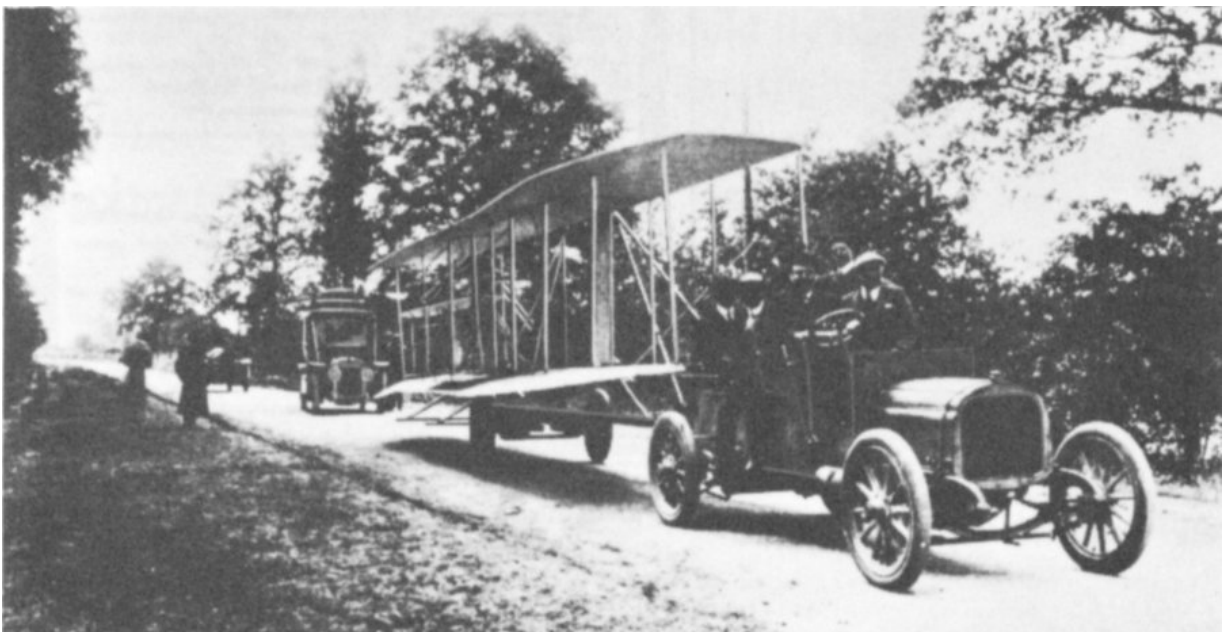
Original patent issued to the Wrights.



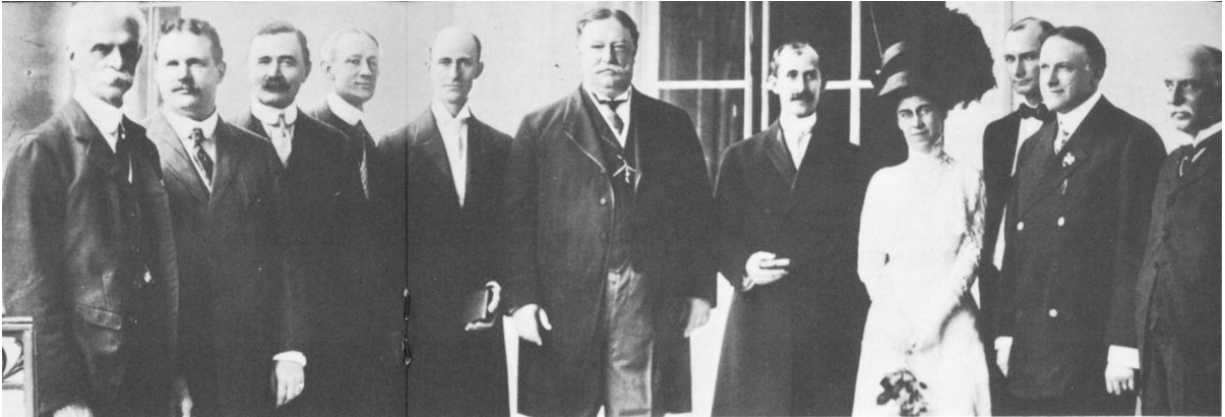
Katherine and Orville Wright aboard ship bound for Europe.



Orville Wright with Thomas A. Edison.



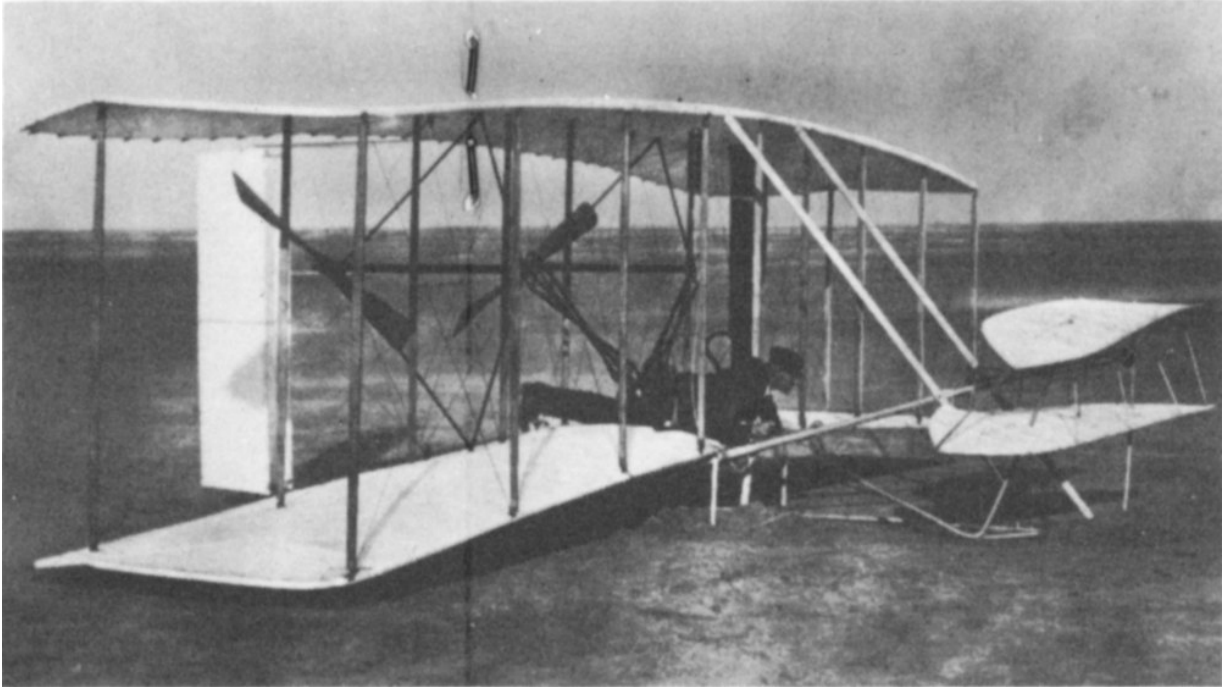
Towing the plane from one field to another at Le Mans, France.



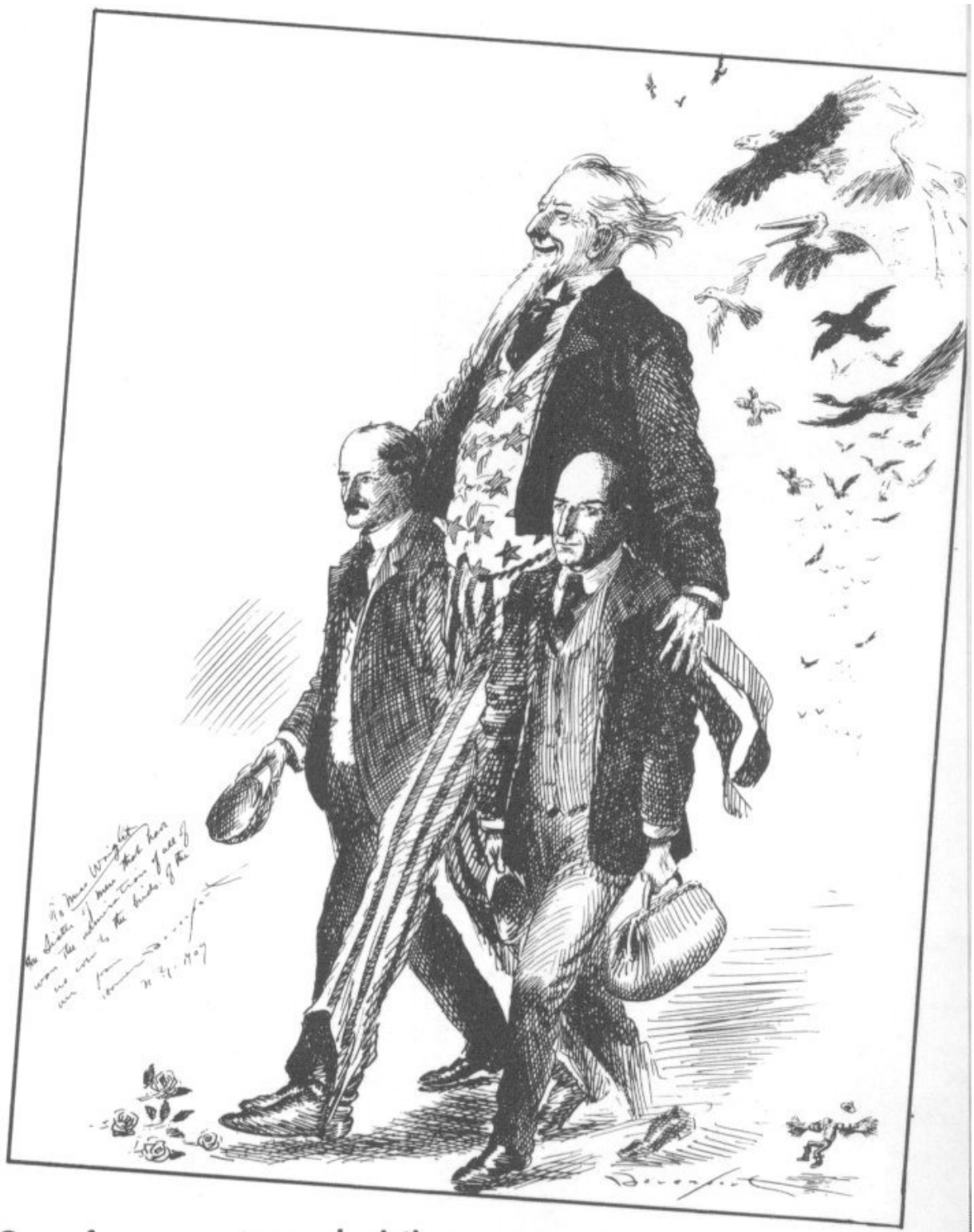
Upon their return from Europe in May 1908, the Wright brothers and their sister, Katherine, were received at the White House by President Taft.



Orville Wright, members of his family and fellow Daytonians at dedication of the Kitty Hawk monument.



The pilot lay prone in early Wright planes.



One of many cartoons depicting the honors which came to the Wright brothers.

He Is Lucky—Page 6

DAYTON HERALD

HOME EDITION

DAYTON, OHIO, THURSDAY, MAY 13, 1909.

TEN PAGES.

PRICE TWO CENTS.

OTHER FORECAST: Day mostly cloudy to sunny and warm; some light to heavy showers.

UNS BOOM, WHISTLES BLOW, CROWD CHEERS WHEN THE WRIGHTS ARRIVE



"OH, THERE'S DADDY," SAYS KATHERINE WRIGHT, AND THEN FATHER AND CHILDREN EXCHANGE GREETINGS—DEAFENING APPLAUSE GREET CONQUERING AVIATORS.

Amid the roar of cannon and screaming of whistles, boys and girls tendered in honor of their wondrous exploits with the aeroplane aboard, Wilbur and Orville Wright and sister Katherine came home today.

Pennsylvania train No. 302, bearing them, arrived at 12:05 this afternoon.

"We are naturally mighty glad to be home again," said Wilbur Wright more as a nod than verbal expression.

"We were much pleased with the work abroad."

"Will you consent to give an aeroplane flight at the reception here next month?" Orville was asked.

"We can't say as to that yet. We have nothing to say, in fact."

Further efforts failed signally to fall short as the patriotic parade moved down the avenue, that characterized the coming of the day.

Edgar Rice and Bill Anderson, secretaries of the War, who were in the city, met the party at the hotel and accompanied them to the reception.

Shortly before 12 the reception committee, headed by Mr. C. Morison, assembled at the Union square, and formed a hollow square. Then three passed through the line of men, each waving a red banner, the venerable father of the Wrights, being the first.

And then came the Wrights.

And entering the carriage at the end of the line, the Wrights, without stopping to the cheering of the multitude, started a series of "off" to the city and home and arrived at the hotel at 1:15.

Between the house and the hotel, the Wrights and the family of the Wrights, who were in the city, were in the city, and at the end of the line.

RANK WITH FULTON, EDISON AND MORSE



WILBUR WRIGHT

ORVILLE WRIGHT

Dayton newspaper reporting the Home-coming Celebration in honor of the Wright brothers.



Orville Wright with Colonel E. A. Deeds on a visit to Wright Field.



Orville Wright with Henry Ford as Dayton honored 35th anniversary of flight.



The Wright home and bicycle shop as they appear today in Greenfield Village, Dearborn, Michigan.

During 1904 more than 100 flights had been made at Huffman Prairie. 15 Of those flights, a complete circle made for the first time on the 20th of September and two flights of three miles each were the most notable. In May of 1905 the Wrights made various improvements in the machine making it much stronger at various points which had proved weak when landing in 1904 flights. The warping of the wings and operation of the tail rudder were made independent of each other and the camber of the wings was changed. The most important development was the addition of two “blinkers” between the surfaces of the front elevator. The purpose of the blinkers was to assist the rear rudder in making a turn. This device was patented and proved quite important for it removed the danger of a tail spin.

The Wright brothers considered their flights of 1905 of great importance, and the 1905 plane proved through performance that it was a greatly improved “flyer.” In a report to the Aero Club of America dated March 12, 1906, they said this:

“The object of the 1905 experiments was to determine the cause and discover remedies for several obscure and somewhat rare difficulties which had been encountered in some of the 1904 flights and which it was necessary to overcome before it would be safe to employ flyers for practical purposes. Toward the middle of September, means of correcting the obscure troubles were found and the flyer was at last brought under satisfactory control. From this time forward almost every flight established a new record.” The last flight was the longest of all, lasting for 38 minutes and 3 seconds and covering $24\frac{1}{5}$ miles. It ended because of exhaustion of fuel. The gas tank, which held only about a gallon, had, through oversight, not been full before the take-off.



One of the many flights made over Huffman Prairie, just east of Dayton on present site of Wright-Patterson airfields.



The Wright Memorial overlooking Huffman Prairie.

THE WRIGHT MEMORIAL

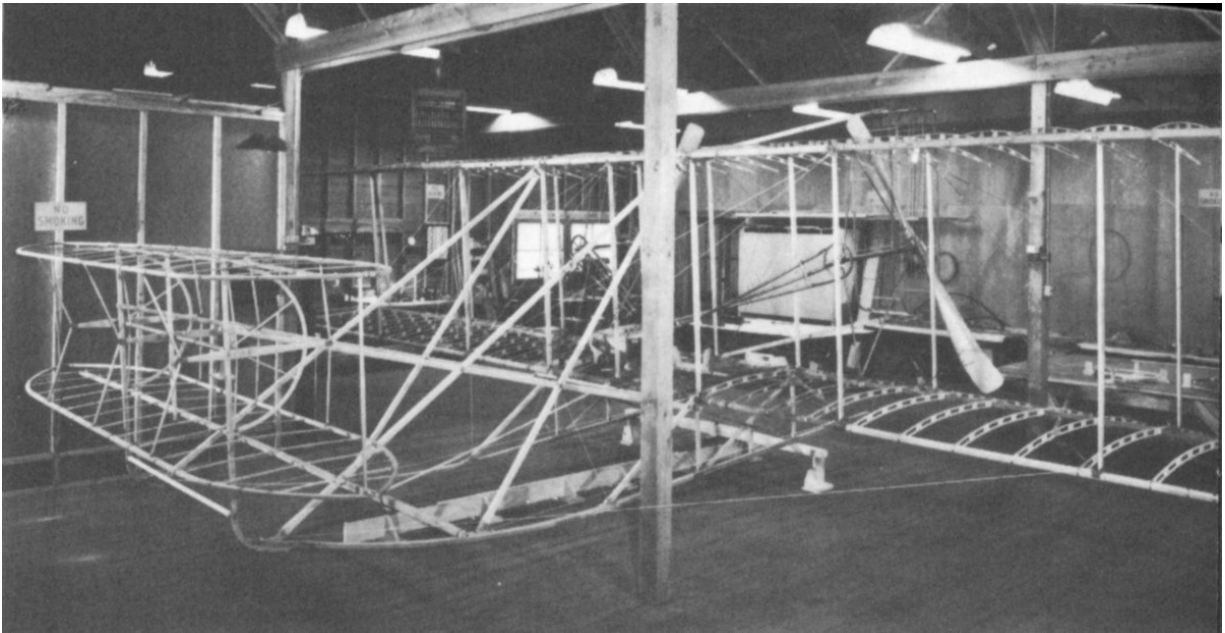
On a hilltop overlooking Huffman Prairie where the Wright brothers accomplished so much, stands this shaft, made of North Carolina pink granite and erected by the citizens of Dayton in their memory. To the east lies Wright Field, the great government air center named in their honor. The principal bronze plaque tells in a few words the story of their great contribution to the progress of mankind. A smaller tablet records the names of those early flyers who were trained by the Wright brothers. In the simplicity of design and the strength of structure, it reflects the characters of the men it honors.

The report to the Aero Club continued, “The 1905 flyer had a total weight of about 925 pounds, including the operator and was of such substantial construction as to be able to make landings at high speed without being strained or broken. From the beginning the prime object was to devise a machine of practical utility, rather than a useless and extravagant toy.... The favorable results which have been obtained have been due to improvements in flying quality resulting from more scientific design and to improved methods of balancing and steering.... The best dividends on the labor invested have invariably come from seeking more knowledge rather than more power.”

The submission of this report was followed by the adoption of a resolution by the Aero Club commending the Wrights upon their accomplishment. This, it might be said, marked official recognition on the part of the public that the Wrights really had flown. Despite numerous flights made in 1904 and 1905, there was considerable skepticism and grave doubts on the part of most people that flights were being made. In fact, the unwillingness of the world to believe that man could fly was one of the ironies of the Wright story. It was many months before the last doubting Thomas was convinced that practical flight had actually been achieved.

Improvements had been made on the 1905 plane, including the engine. In 1908 the plane was taken to Kitty Hawk for further tests. After several

successful flights, an accident occurred which so badly damaged the plane that it was dismantled and stored there in frame hangars. Over the years parts of the plane were given to several museums and others were acquired by residents of the area as mementoes. The engine, the propellers and other parts were shipped back to Dayton.



The restored 1905 aeroplane in process of reconstruction. Engineers and others who inspected the plane during its rebuilding, marveled at the craftsmanship reflected in its original construction.

When it was decided to reconstruct an early Wright plane for Carillon Park, the first thought was that it should be a replica of the Kitty Hawk, which of course would have been accurate in appearance but would have contained no original parts.

Orville Wright himself suggested that if the original parts of the 1905 plane could be brought together, a plane which could truly be called a restored Wright aeroplane could be built. An exhaustive search was begun and with the co-operation of the museums and the residents of Kitty Hawk, many of the original parts were secured. Orville Wright located the original drawings and supervised much of the reconstruction. His death occurred shortly before the plane was finished.

At least 60 per cent of the parts in the plane are original. These include the engine, the chain guides, control levers and pilot's cradle, the propellers, the greater part of the wing structure as well as some of the front rudder struts. Construction of the plane was supervised by Mr. Harvey D. Geyer, an early employee of the Wrights, who was uniquely fitted for this responsibility and who, in contributing his services, has done much to perpetuate the achievements of the Wrights in their home city. As does the original Kitty Hawk in the Smithsonian Institution, this restored plane will, for generations to come, help to tell the story of the genius of the Wrights.

WILBUR WRIGHT FLIES UP THE HUDSON RIVER

From Governors Island to Grant's Tomb and Back.

FLYING IN AIR WILL BE POPULAR

Wright in Air Seven Minutes

Dayton Aerialist Makes Another Successful Flight, Reaching Highest Level of Sixty Feet—Is Inconvenienced by Large Crowd of Spectators.

avorable conditions he has experienced. The wind, which helped him in the morning trial, became gusty, and the aeronautist deemed it advisable not to attempt it again. But he sailed around close to the ground, however, with the same perfect ease and skill with which the previous experiment was carried out. Mr. Wright today said to The Associated Press that he was greatly inconvenienced by the number of per-

Wright's Aeroplane Ascends Like Bird; Control Is Perfect

Wright Airship Makes Ten Minute Flight



ORVILLE WRIGHT'S FLIGHT DELIGHTS OFFICERS OF ARMY

ASPECT OF WRIGHT AEROPLANE.



WILBUR WRIGHT FLIES MORE THAN HOUR AND HALF, REMARKING SIMPLY: "THIS WILL CHEER ORVILLE UP A BIT."

droplane, in full flight. Yet Made in France, Circling Control of the Aeronaut, Satisfied.

the American Ambassador and Aero-Club Officials He Easily Breaks World's Record.

VINDICATES HIS MACHINE.

His Mettle to Prove Recent Mishap Merely Accidental, His Object to Attain.

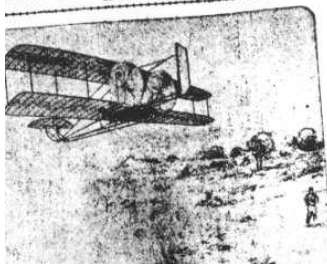
DAYTON FLYER MAKES FIRST PERFECT TEST

NEW YORK HERALD.

SUNDAY, SEPTEMBER 6, 1908. NINETY-EIGHT PAGES. THE NEW YORK HERALD. Wright Airship, Nearly 20 Minutes in Air, Flies 15 Miles, Approaching Within Few Seconds Farman's Best Record

WORLD'S FLYING RECORD MADE BY AVIATOR WRIGHT

Encircles Parade Grounds at Ft. Myer 56 Times and Remains in the Air 57 Minutes and seconds—Early Ascension Made in the Presence of Only 25 Spectators Smashes All Aero-Records, Easily Exceeding DeLagrange's Recent Remarkable Flights.



WRIGHT, IN 73-MILE FLIGHT, BREAKS ALL WORLD RECORDS

Throng at Le Mans, France, Cheers American, Who Sails Two Hours in Air.

IN FACE OF ICY WIND

Achievement Wins for Him \$5,000 Michelin Cup—Descends and Flies Again

Le Mans, France, Sept. 21.—Wilbur Wright, the American aviator, today, in the presence of a throng of 20,000, used through the air to his aviator for two hours and after making a new world

FRANCE HONORS WILBUR WRIGHT MAGNIFICENTLY

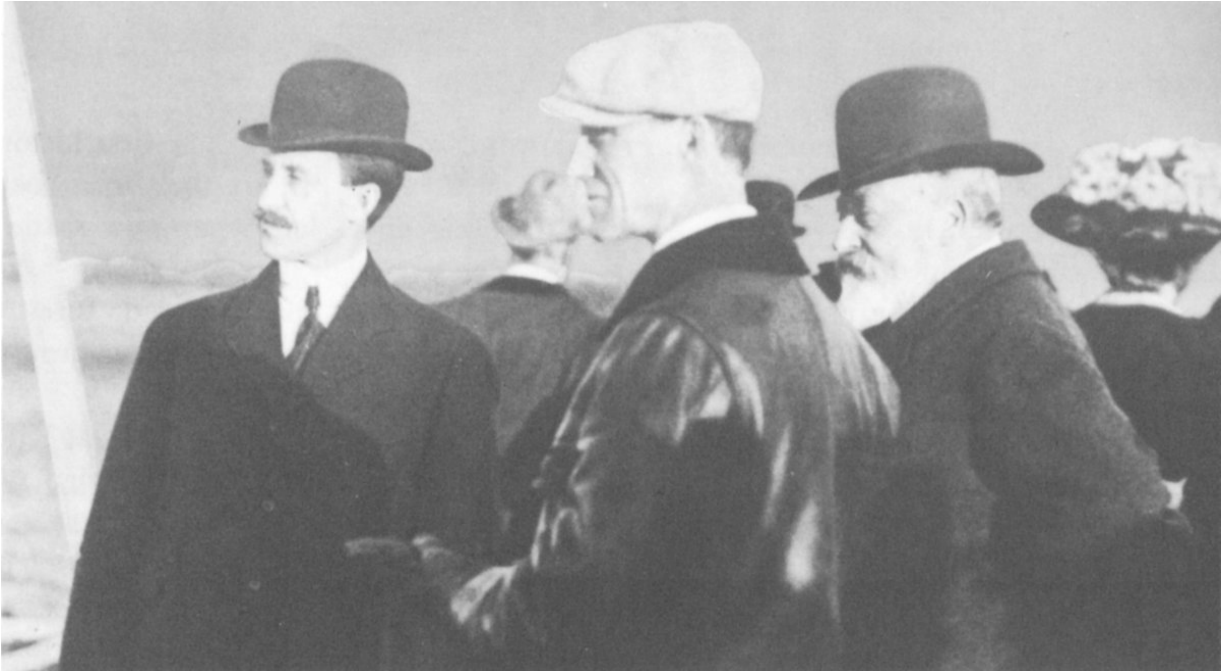
Statesmen and Scientists of Republic Pour Forth Most Extravagant Praise for Aviator.

TWO GOLD

KING VICTOR EMMANUEL WATCHES FLIGHTS BY MR. WILBUR WRIGHT



Recognition ... at home and abroad

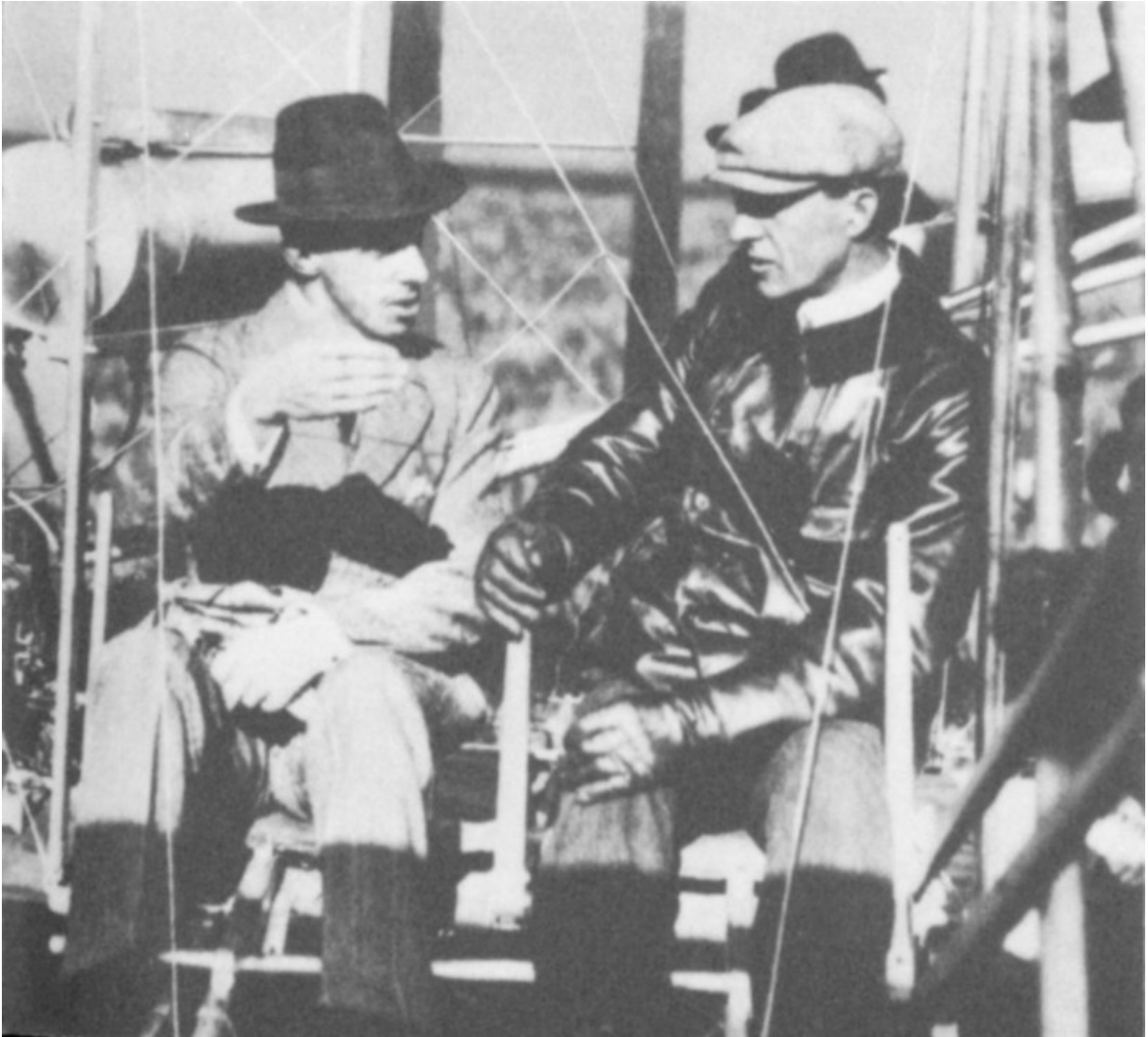


King Edward VII visits Wright brothers during flights at Pau, France.

A world that has convinced itself something cannot be done, yields slowly to the realization that the “impossible” has been achieved. When the Wrights approached their own government with the suggestion that their invention might be useful for scouting purposes their proposal evoked no interest. Actually, appreciation of the implications and possibilities of the new device came more quickly from Europe than America. England and France were among the first to seek information on the machine that had so thoroughly proved its ability to fly. As early as 1905 a member of the French military had at least made unofficial inquiry as to the cost of a plane, but for a time this led to nothing.

In 1907 the United States government realized that the Wrights had proved the practicability of flying. The Signal Corps drew up specifications and asked for bids. The Wrights offered to build a test plane for \$25,000. Their bid was accepted in February, 1908. Three months later they signed a

contract with a French syndicate to sell or license the use of the plane in France. The Wrights were now in the international picture.



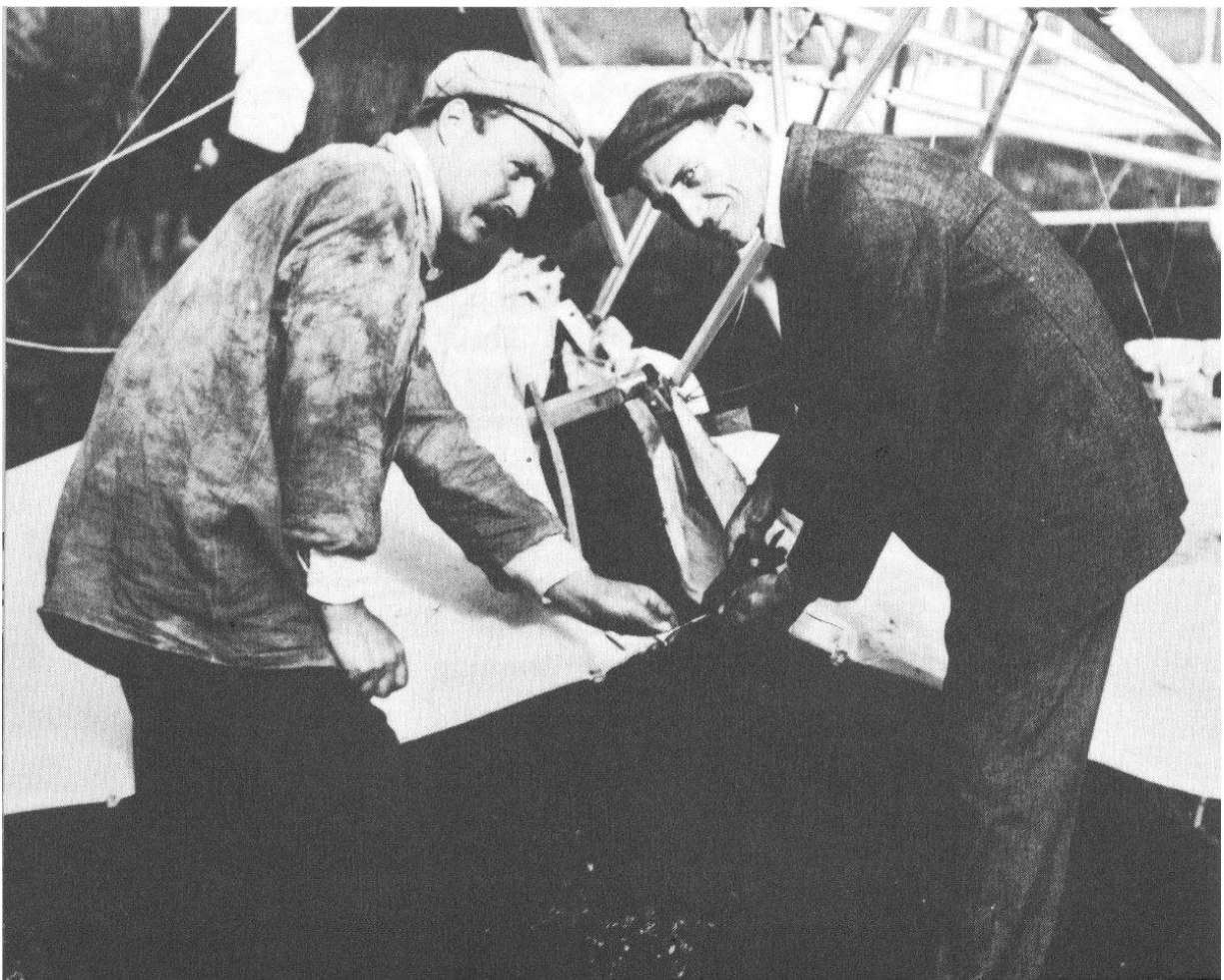
King Alfonso XIII of Spain was keenly interested in flying, but promised his family that he would not make flight.

About this time the Wrights, always seeking better performance, made a notable improvement in their plane. In their first historic flight and during the experiments on Huffman Prairie, they rode “belly buster” just as a boy does when coasting on a sled. They now made a different arrangement of levers which enabled them to sit up while piloting the plane. A seat for a passenger was also provided. Interestingly enough, recent

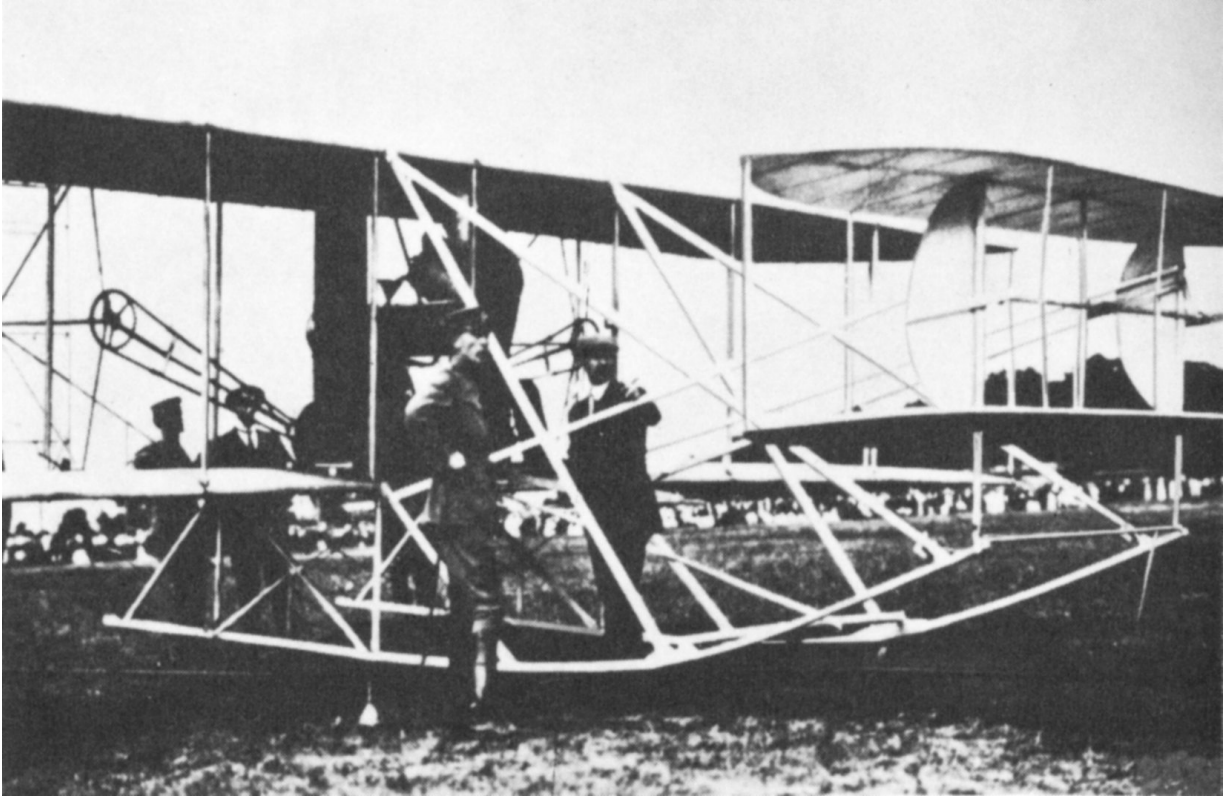
experiments with high-speed planes have brought some return of the prone position for the pilot.

On May 14, 1908, newspaper men saw a history-making flight at Kitty Hawk. The remodeled 1905 machine under perfect control carried two men. Flights for the army followed in September and the last trace of skepticism disappeared. Unfortunately, on the last flight Lieutenant Selfridge, the passenger, was killed and Orville severely injured.

The year 1908 was notable in the saga of the Wrights. Wilbur made a series of flights abroad that not only won all observers but aroused wide interest and admiration throughout Europe. His quiet demeanor, his unassuming modesty and his proved skill, stirred the popular imagination. The French exalted him to the status of a hero. The great of the world flocked to meet him and see him fly. They included King Edward VII of England, King Alfonso XIII of Spain, and the Dowager Queen Margherita of Italy. Invitations to fly came from Rome and Berlin. In Rome King Victor Emmanuel watched him fly.



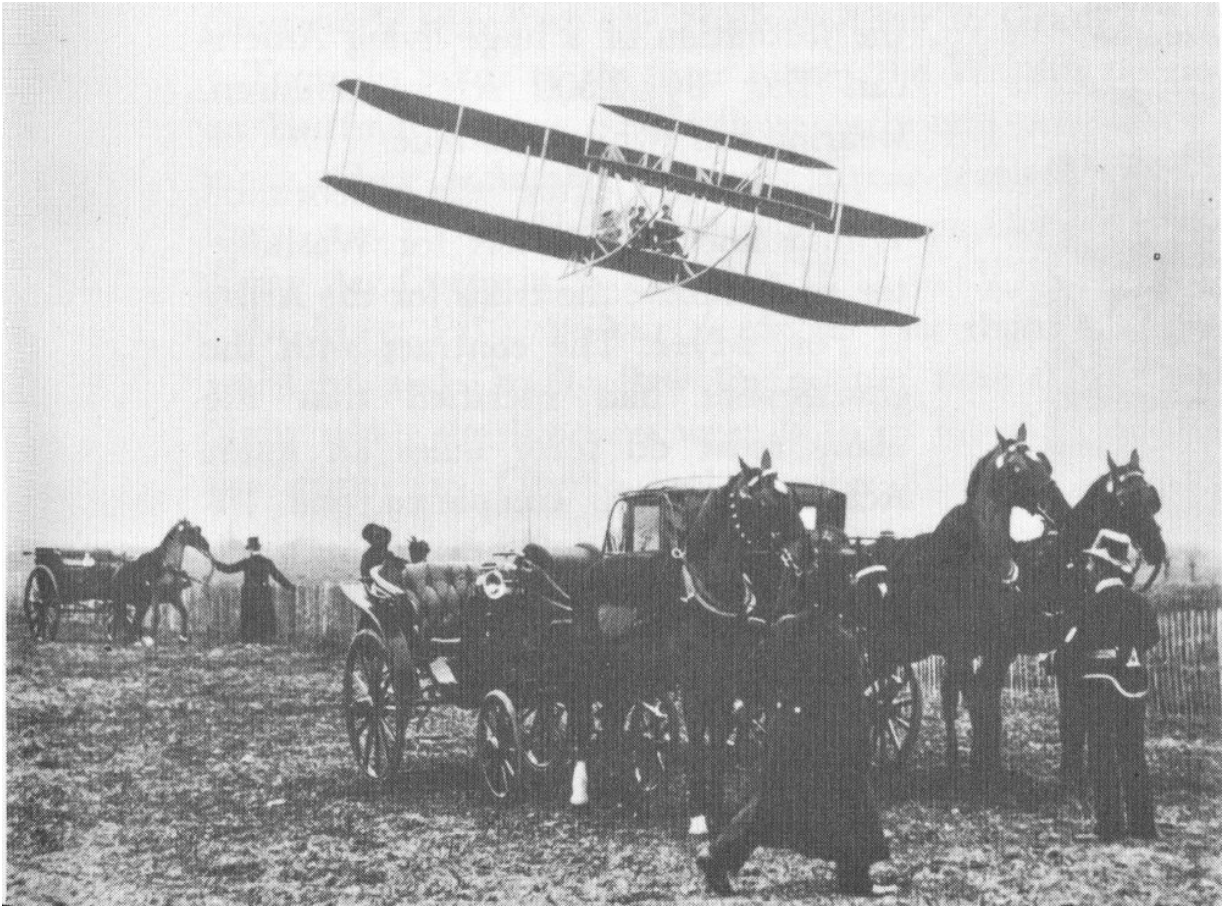
Wilbur, at right, in characteristic pose—making a repair in France.



Orville Wright with army officer during highly successful flights at Fort Myer, Virginia.

In December, 1908, Orville and his sister, Katherine, went to Europe to join Wilbur. The weather at Le Mans where Wilbur had been flying became unsuitable for further flights and operations were transferred to Pau in southern France. Here Orville and Katherine joined Wilbur. Many flights were made and many distinguished visitors came to see the modern miracle of human flight.

Honors were heaped upon the Wrights. They received among many other distinctions, the gold medal of the Aeronautical Society of Great Britain and the Aero Club of the United Kingdom. The French Aero Club of the Sarthe gave them a bronze trophy. Later the Aero Club of America bestowed medals on the flyers. A few weeks afterward, President Taft received the Wrights at the White House and the brothers returned to Dayton where a tumultuous welcome awaited them.



Wilbur flies low over spectators' carriages at Pau, France.

Welcome home!



Main Street and Third on day of great Homecoming Celebration, June 17, 1909.

Probably nothing stirred the Wrights quite so deeply as their welcome when they returned to Dayton from their foreign triumphs. The “homecoming” lasted two days, June 17 and 18, 1909. Whistles blew, bands played, bells rang, men, women and children paraded. During the celebration practically all business in Dayton was suspended.



Orville and Wilbur on second day of Dayton demonstration.

Wilbur and Orville rode to the celebration in a carriage with their old friends, Ed Sines, boyhood newspaper partner of Orville, and Ed Ellis, a long-time friend of Wilbur. The Wrights reviewed a parade in their honor and in the evening witnessed a spectacular display of fireworks. The celebration continued the next day when one of the features was the formation of a huge living American flag by 2,500 school children, wearing red, white and blue. Immediately after the celebration Wilbur and Orville left for Washington to complete the trials for the Army at Fort Myer. The contract with the government had specified that the plane must do forty miles an hour. Actually, Orville completed one 10-mile flight in 14 minutes at approximately 43 miles per hour. The Wright plane was accepted by the Army at the conclusion of these tests.



Michelin Trophy awarded to Wright brothers for achievements in France.



{Medal from the Aero Club of America.}



This trophy from the Aero Club of Sarthe, France, was placed in niche in Wright home.

Immediately after the flights at Fort Myer, Orville and Katherine left for Germany. His purpose was to train pilots for the German company which had been organized. He made many flights on that trip, some of them witnessed by members of the royal family and on one of which the Crown Prince was a passenger. On one he raised the world's altitude record from 100 meters to 172 meters, roughly 550 feet. Shortly thereafter he flew for one hour, thirty-five minutes and forty-seven seconds with a passenger, thereby establishing a new world's record for a flight with a passenger.

While Orville was in Germany in 1909, Wilbur was making spectacular flights around New York. In one of these he flew 21 miles from Governor's Island up the Hudson River to Grant's Tomb and back.

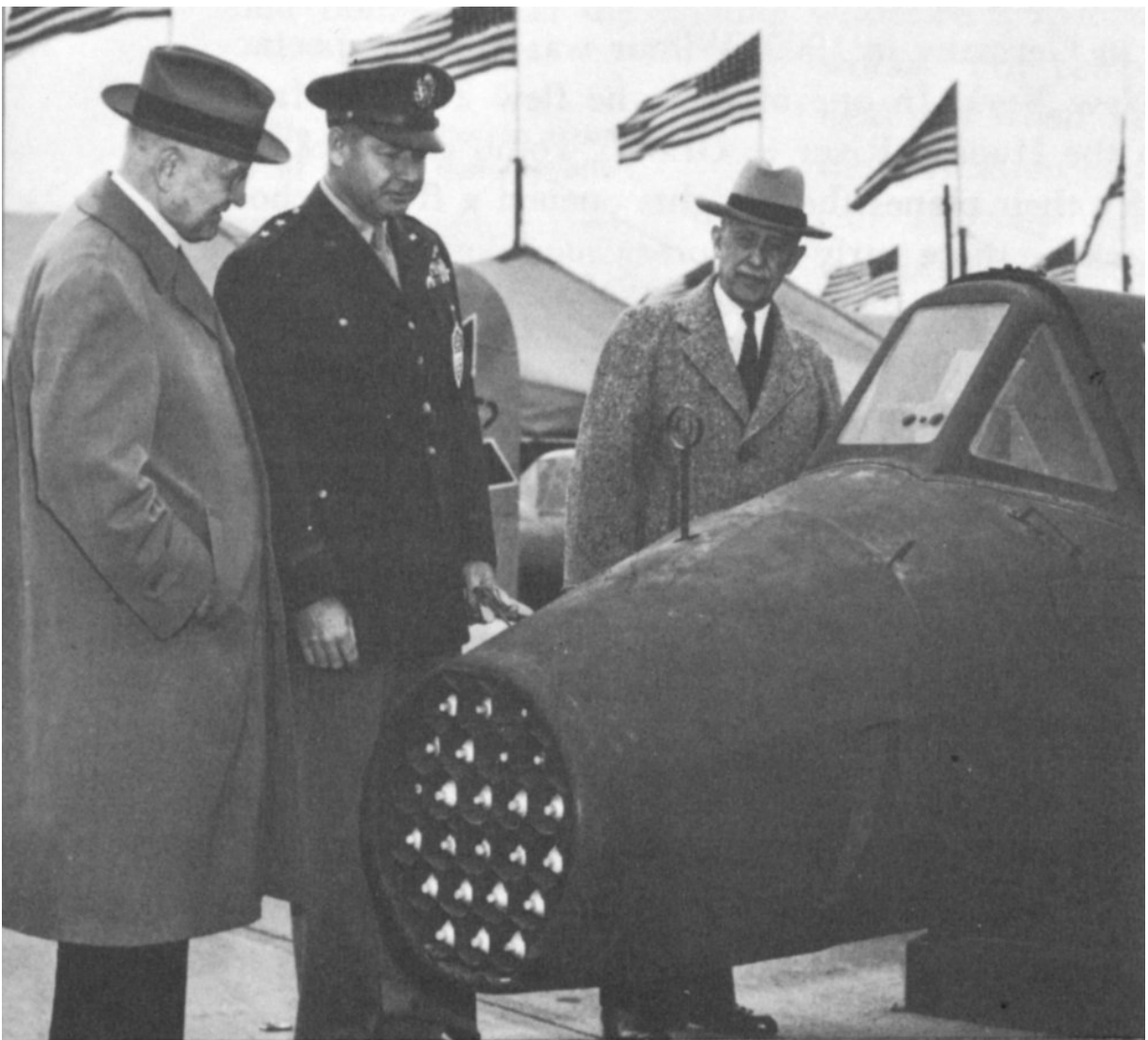
To train pilots to fly their planes the Wrights opened a flying school on Huffman Prairie where those early and precarious flights had been made. Here a notable group of flyers received their training. One of them was Henry H. Arnold who became Commanding General of the Army Air Corps in World War II.

In May, 1910, Wilbur made his last flight as pilot. Shortly afterward he and Orville flew for a brief time together. It was the only flight when the brothers were both in the air at the same time. Later the same day Orville took up his 82-year-old father. In the spirit of the Wrights the Bishop's only comment was ... "Higher! Higher!" Orville's final flight as pilot was made in 1918 from South Field near Dayton.

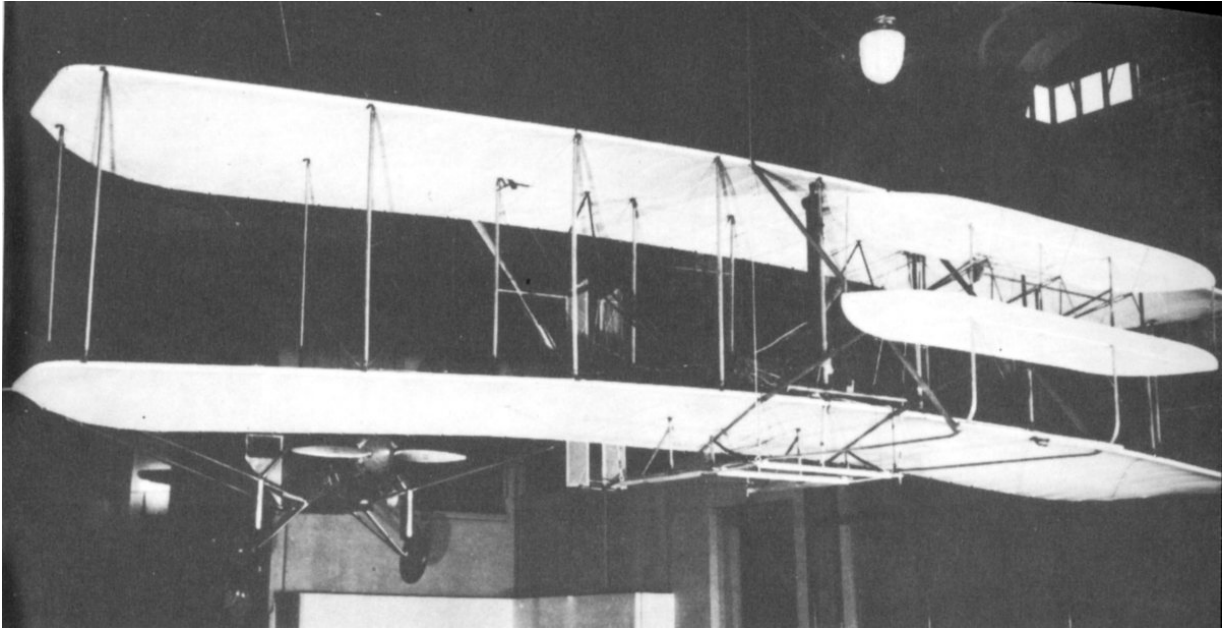


Orville Wright meeting with members of National Advisory Committee for Aeronautics.

After an illness of three weeks, Wilbur Wright died on May 30, 1912, in his forty-fifth year. The whole world mourned him. Thus, in the prime of life, with a record of achievement privileged to few, passed a notable figure in American creative history. Orville Wright survived his brother for 36 years, passing away January 30, 1948. Throughout his life he maintained his active interest in aviation, was a life member of the National Advisory Committee for Aeronautics and a frequent and honored visitor to Wright Field, the great Air Force research center named in honor of the Wright brothers.



Always interested in new developments, Orville Wright visits Wright Field.



The “Kitty Hawk,” Smithsonian Institution, Washington.

The Wright brothers have been honored by many nations. Medals, trophies, monuments tell in part, at least, the story of their great achievement. The original Kitty Hawk aeroplane holds the place of honor in the aeronautical exhibit of Smithsonian Institution, Washington. A replica of the Kitty Hawk in the Science Museum at South Kensington, London, speaks for the British nation in honoring the Wrights. Monuments have been erected at Kitty Hawk, N. C., at Le Mans, France, and at Dayton. And now Wright Hall with its restored 1905 plane takes its place as one of the efforts of a grateful world to honor one of man’s greatest achievements.



Wright Memorial, Kitty Hawk, N. C.



Monument to Wrights, Le Mans, France.



Wright Hall in Carillon Park houses restored 1905 Wright aeroplane.

On the walls of Wright Hall is inscribed this tribute to the achievements and to the personal character of two great Americans:

In honored memory of Wilbur and Orville Wright, citizens of Dayton and of the world. Through original research, the Wright brothers acquired scientific knowledge and developed theories of aerodynamics which, with their invention of aileron control, enabled them in 1903 to build and fly, at Kitty Hawk, the first power-driven, man-carrying aeroplane capable of flight.

Their further development of the aeroplane gave it a capacity for service which established aviation as one of the great forward steps in human progress.

As scientists, Wilbur and Orville Wright discovered the secret of flight. As inventors, builders and flyers, they brought aviation to the world.

Their courage, perseverance and ability are comparable only to the magnitude of their achievement. The aeroplane will stand for all time as one of those few truly great inventions which have shaped the life and destiny of man.

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